

• OFFICIAL SYSTEM SPECIFICATION • PRODUCTION BASELINE V2.4

Travels
Pro:

Enterprise Travel & Multi-Modal
Mobility OS

Exhaustive technical handbook, domain architecture, module mechanics, logical & UI flows, complete data dictionary, legal compliance framework, financial unit economics, and engineering roadmap.

DOCUMENT CLASSIFICATION	REGULATORY COMPLIANCE	ENGINEERING CORE
Master Technical Blueprint	DPDP Act 2023 • IT Act 2000	Next.js 16 • Turbopack • Prisma 7
RELATIONAL SCALE	AUTOMATED VERIFICATION	PRODUCTION ENDPOINTS
70 Models • 46 Enums • 28 Migrations	212 / 212 Tests Passed (100%)	47 UI Routes • 119 API Endpoints (152 Operations)

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01 Executive Overview, Problem Analysis & Statutory Governance

Executive Statement: Travels Pro transforms fragmented, informal experiential travel into a verified, secure, and multi-modal digital marketplace. By integrating **Group Tours**, **Hyperlocal Activities**, and **Self-Drive Car Rentals** under a unified zero-trust identity and escrow umbrella, it creates institutional trust for both travelers and supply operators.

1.1 Market Opportunity & Core Industry Deficiencies

Experiential travel and localized adventure tourism in India and Southeast Asia represent an explosive, rapidly expanding market sector. However, the sector is plagued by systemic operational vulnerabilities that inhibit consumer confidence and host scalability:

FAILURE VECTOR	TRADITIONAL MARKET PITFALLS	TRAVELS PRO ENGINEERED ARCHITECTURE
Host Legitimacy & Trust	Unverified operators advertising on Instagram/WhatsApp without legal registration, emergency medical training, or valid vehicle commercial permits.	6-Stage Verified KYC: Mandatory legal identity verification, government ID proof validation (Aadhaar/PAN regex), live biometric selfie with ID matching, and <code>HOST_SAFETY_AGREEMENT</code> compliance.
Data Privacy & PII Leakage	Sensitive identity proofs (Aadhaar cards, passports) exchanged over unencrypted messaging apps without retention or disposal controls.	AES-256-GCM Vault: Zero plain-text storage of bank accounts or tax IDs. Stored with initialization vectors and tamper-evident authentication tags.
Vehicle Custody Friction	Undocumented pre-existing scratches and vehicle defects trigger arbitrary security deposit forfeiture and acrimonious disputes upon return.	Two-Stage Digital Custody: Odometer, fuel level, and 4-angle exterior/interior photos cryptographically logged and countersigned at pickup and return.
Financial Escrow & Refunds	Operators demand 100% upfront UPI bank transfers. In cases of cancellation or substandard delivery, travelers have zero recourse.	Milestone Escrow Clearing: Customer payments are held in escrow and released to host accounts only post-trip completion. Transparent tiered refund rules.
Community & Coordination	Tour groups communicate in chaotic public WhatsApp groups with zero privacy protection, exposing traveler phone numbers to harassment.	Trip Circles: Authenticated, private Socket.IO chat rooms scoped strictly to confirmed booking codes with real-time moderation and SOS broadcasting.

1.2 Statutory Legal Baseline: India DPDP Act 2023 Compliance

Travels Pro was engineered from the ground up to comply with the **Digital Personal Data Protection Act, 2023 (DPDP Act 2023)** and the **Information Technology (Reasonable Security Practices and Procedures and Sensitive Personal Data or Information) Rules, 2011**. Key compliance tenets include:

- Unbundled Explicit Clickwrap Consent:** Pre-checked checkboxes are strictly prohibited across all signup, login, and checkout forms. Users must voluntarily click distinct checkboxes for Terms of Service, Privacy Policy, and Safety Guidelines.
- Tamper-Evident Consent Logging:** Every consent event is persisted into an immutable `PolicyConsent` record capturing `userId`, `policyId`, `policyVersion`, `ipAddress`, `userAgent`, and `context` (e.g. `SIGNUP`, `HOST_KYC`, `BOOKING`).
- Dynamic Force Re-Consent Engine:** When administrators publish an updated policy version (e.g. v1.0 → v1.1), the authentication gateway intercepts subsequent login requests, sets `reconsentRequired: true`, and mounts a blocking `PolicyConsentModal` that prevents application navigation until consent is granted.
- Data Principal Rights & Erasure:** Implements self-service data export and account erasure endpoints adhering to statutory 30-day fulfillment windows.

1.3 The Five Standard Platform Policies

POLICY TYPE	TITLE	KEY STATUTORY CLAUSES & ENFORCEMENTS
<code>TERMS_OF_SERVICE</code>	Terms of Service & User Agreement	Marketplace intermediary protections under Section 79 of IT Act 2000, binding digital contracts, payment escrow terms, and dispute arbitration under Arbitration and Conciliation Act, 1996.
<code>PRIVACY_POLICY</code>	Privacy Policy & Data Protection	Purpose specification, lawful processing grounds under DPDP Act 2023, cookie usage, data retention schedules, and grievance redressal officer contact disclosures.
<code>HOST_SAFETY_AGREEMENT</code>	Host Safety Agreement & Code of Conduct	Mandatory statutory permits, pre-trip vehicle roadworthiness inspections, emergency first aid kit

POLICY TYPE	TITLE	KEY STATUTORY CLAUSES & ENFORCEMENTS
		maintenance, certified guide requirements, and zero tolerance for harassment.
TRAVELER_SAFETY_POLICY	Traveler Safety Guidelines	Fitness declarations, compliance with certified host safety directives, substance and alcohol prohibitions during high-risk activities, and Leave No Trace environmental guidelines.
CANCELLATION_POLICY	Standard Cancellation & Refund Policy	Unambiguous tiered refund boundaries (>48h: 100%, 24–48h: 50%, <24h: non-refundable), host cancellation penalties, and force majeure weather exemptions.

02 Enterprise System Architecture & Cloud Infrastructure

Travels Pro operates as a decoupled, multi-tier system combining an optimized Next.js 16 frontend shell, a high-throughput REST API gateway, a serverless PostgreSQL database with connection pooling, in-memory Redis acceleration, and an event-driven Socket.IO messaging cluster.

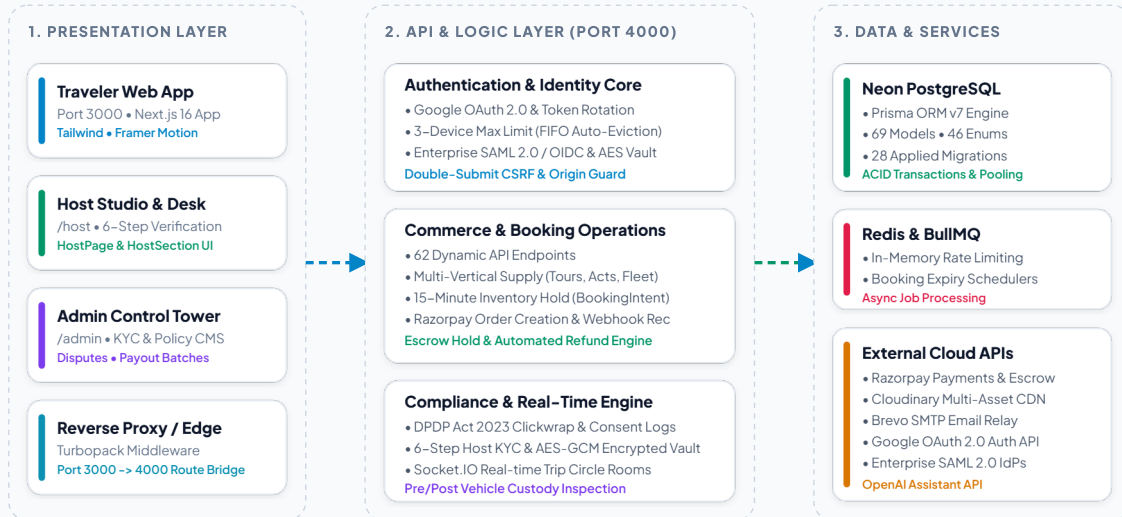


FIGURE 2.1: TRAVELS PRO ENTERPRISE DISTRIBUTED SYSTEM TOPOLOGY & INFRASTRUCTURE MAP

2.1 Component Deep-Dive & Architectural Roles

1. Presentation Shell (Port 3000)

Built with **Next.js 16 App Router**, React 19, and Turbopack. Renders 47 production routes utilizing Server-Side Rendering (SSR) for SEO-sensitive discovery pages and Client-Side Hydration (CSR) for rich stateful dashboards. Employs **HostUI** design tokens, Framer Motion transitions, and Radix UI primitives.

2. Headless API Gateway (Port 4000)

Hosts 119 modular REST endpoints (152 HTTP operations) built on the Node.js server runtime. Implements strict Zod schema validation, origin verification, double-submit CSRF prevention, rate limiting, and centralized error handling with sanitization to prevent stack trace leakage.

3. Persistence Layer (Neon PostgreSQL)

Utilizes **PostgreSQL 16** hosted on Neon Serverless, managed via **Prisma ORM 7** with 28 applied sequential migrations. Employs PgBouncer transaction-mode connection pooling to handle high-concurrency bursts across 70 relational models and 46 strict enums.

4. Real-Time & Asynchronous Layer

Socket.IO cluster handling sub-50ms message propagation for Trip Circles, authenticated via session cookies. **Redis & BullMQ** workers manage 15-minute booking hold expirations, notification email relays, and webhook event processing.

2.2 Cryptographic Security Vault & Authentication Engine

- **AES-256-GCM Vault:** All sensitive host financial records (bank account numbers) and identity credentials are encrypted at rest using AES-256-GCM. Ciphertext strings are stored in the format `${iv}:${ciphertext}:${authTag}`. Tampering with either the tag or ciphertext throws an immediate cryptographic authentication error.
- **JWT HS256 with Device ID Binding:** Issues short-lived access tokens (15-minute TTL) and rotating refresh tokens (7-day TTL) stored in secure, HttpOnly, SameSite=Lax cookies. Each token embeds a unique `deviceId` fingerprint.
- **FIFO Device Eviction Algorithm:** Accounts are limited to a maximum of 3 concurrent active devices. Upon a 4th login, the oldest device record in `UserDevice` is automatically marked as `EVICTED`, terminating older sessions without blocking users with disruptive 409 errors.
- **Mutation Origin & CSRF Guard:** Mutating HTTP methods (`POST`, `PUT`, `PATCH`, `DELETE`) validate incoming `Origin` and `Referer` headers against trusted whitelist domains and enforce cryptographic double-submit cookies.

03 Traveler Persona Journey: 14 Screen Profiles & State Machines

The traveler lifecycle is designed for zero-friction discovery, dynamic search filtering, instant inventory hold locking, statutory clickwrap compliance, and real-time social trip coordination. Below is the exhaustive breakdown of all 14 screens and their client/server state machines.

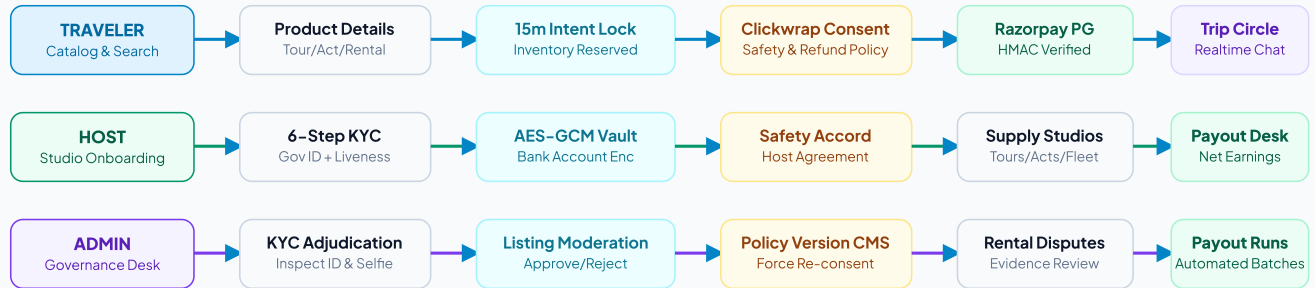


FIGURE 3.1: PERSONA JOURNEY FLOWCHART ACROSS TRAVELER, HOST & ADMIN LIFECYCLES

3.1 Exhaustive Catalog of Traveler Profile Screens (Screens 1 to 14)

Screen 1: Marketplace Discovery Portal (/)

Discovery & Landing Phase

Primary Function: Omnichannel travel discovery hub presenting curated multi-modal travel options (Tours, Activities, Vehicle Rentals) tailored to the traveler's detected location and search preferences.

UI COMPONENT / WIDGET	USER INTERACTION & BEHAVIOR	SERVER & DATABASE EXECUTION
Hero Global Search Bar	Traveler inputs destination keyword (e.g. "Manali", "Goa"), selects vertical tab, and picks target date ranges.	Debounces query by 300ms, hits <code>/api/location/gps</code> for geocoding, queries PostgreSQL full-text search indexes on <code>city</code> , <code>title</code> , and <code>tags</code> .
Vertical Category Switcher	Interactive toggle switching between "Group Tours", "Hyperlocal Activities", and "Car & Bike Rentals".	Client router pushes state without full reload; triggers prefetching of respective catalog datasets.
Trending Destinations Carousel	High-res destination cards with live starting prices, active tour counts, and traveler review badges.	Redis-cached aggregation of top-booked destinations over the trailing 30 days (cache TTL: 1 hour).
Verified Host Spotlight	Displays top-rated hosts with verified KYC badges, total completed expeditions, and average ratings.	Queries <code>Host</code> joined with <code>UserProfile</code> where <code>isVerified: true</code> and <code>averageRating >= 4.8</code> .

Screen 2: Universal Search & Catalog Grid (/tours , /activities , /car-rental)

Exploration & Filtering Phase

Primary Function: High-performance faceted catalog supporting multi-criteria filtering, price sorting, interactive maps, and real-time availability badges.

UI COMPONENT / WIDGET	USER INTERACTION & BEHAVIOR	SERVER & DATABASE EXECUTION
Faceted Filter Sidebar	Sliders for Price Range (₹500 - ₹50,000), Duration (1-14 days), Difficulty (Easy, Moderate, Extreme), and Group Size.	Generates dynamic Prisma <code>where</code> clauses with compound indexes: <code>[city, category, status]</code> and <code>[price]</code> .
Interactive Map Pin View	Map view plotting meeting coordinates and tour starting points with interactive price hover cards.	Executes spatial bounding-box query returning geo-coordinates within traveler viewport.
Sort Order Dropdown	Options: "Most Popular", "Price: Low to High", "Price: High to Low", "Highest Rated", "Upcoming Departures".	Applies SQL <code>ORDER BY</code> on indexed columns: <code>averageRating DESC</code> , <code>price ASC</code> , or <code>createdAt DESC</code> .

Screen 3: Tour Detail & Itinerary Matrix (/tours/[slug])

Evaluation & Consideration Phase

Primary Function: Comprehensive multi-day itinerary presentation with day-by-day activities, meal inclusions, accommodation tiers, safety guidelines, and live departure batch selector.

UI Component / Widget	User Interaction & Behavior	Server & Database Execution
Day-by-Day Accordion	Collapsible cards for each expedition day showing morning/afternoon/evening schedule, elevation gains, and meals.	Hydrates <code>TourItineraryDay</code> sorted by <code>dayNumber ASC</code> ; includes photo galleries and coordinate markers.
Departure Batch Calendar	Interactive calendar displaying specific departure dates, total capacity, and remaining available seats.	Fetches <code>TourBatch</code> where <code>startDate > now()</code> and <code>status = OPEN</code> , calculating real-time remaining capacity.
Host Trust & Credential Card	Host avatar, badge verification level, response time, completed tours counter, and traveler bio.	Loads host profile from <code>Host</code> , <code>UserProfile</code> , and aggregated rating counts.
Inclusions / Exclusions Matrix	Checkmark list of covered costs (Guide, Gear, Meals, Permits) vs excluded costs (Personal expenses, Insurance).	Stored as PostgreSQL text arrays <code>included String[]</code> and <code>excluded String[]</code> .

Screen 4: Tour Checkout & Traveler Roster (/tours/[slug]/checkout)

Reservation & Booking Intent Phase

Primary Function: Traveler roster data entry, emergency contact collection, 15-minute inventory hold locking, and statutory clickwrap consent.

UI Component / Widget	User Interaction & Behavior	Server & Database Execution
15-Minute Countdown Lock	Visual countdown banner: "Inventory held for 14:59. Complete checkout to secure your seats."	Backend creates <code>BookingIntent</code> with <code>expiresAt = now() + 15m</code> , decrementing <code>availableSlots</code> .
Traveler Roster Form	Input rows for each traveler: Full Legal Name, Age, Gender, Dietary Restrictions, Government ID number.	Validated via Zod schema; staged for creation of <code>TourTraveler</code> records upon payment confirmation.
Emergency Contact Input	Mandatory name, relationship, and 10-digit mobile number for off-grid safety protocols.	Stored on <code>TourBooking.emergencyContact</code> ; accessible to verified tour guides 24 hours prior to departure.
Statutory Clickwrap Modal	Unchecked checkboxes for <code>TRAVELER_SAFETY_POLICY</code> and <code>CANCELLATION_POLICY</code> with inline view modals.	Checkout button remains strictly disabled. Checking boxes prepares payload for <code>PolicyConsent</code> audit logging.

Screen 5: Hyperlocal Activity Slot Selection (/activities/[slug])

Single-Day Activity Booking Phase

Primary Function: Slot-based reservation interface for single-day adventures, workshops, and sports with real-time session capacity counters.

UI Component / Widget	User Interaction & Behavior	Server & Database Execution
Date & Time Slot Grid	Horizontal date carousel with vertical time-slot pills (e.g. "09:00 AM", "01:30 PM", "04:00 PM") showing available spots.	Queries <code>ActivitySlot</code> for selected date, returning <code>startTime</code> , <code>availableSlots</code> , and <code>price</code> .
Participant Counter & Add-ons	Counter incrementing guest count with live price calculation and optional rental gear add-ons (e.g. wetsuit, helmet).	Validates party size against <code>groupSizeMax</code> ; calculates subtotal, taxes, and service fees in real time.
Instructor & Gear Disclosures	Safety equipment certification details, certified instructor credentials, and minimum age/fitness prerequisites.	Loads activity safety metadata and generates prerequisite waiver acknowledgment prompt.

Screen 6: Vehicle Rental Custody Terms & License Upload (/car-rental/[slug])

Fleet Mobility Reservation Phase

Primary Function: Self-drive vehicle reservation interface enforcing statutory driving license verification, security deposit calculation, and fuel policy selection.

UI COMPONENT / WIDGET	USER INTERACTION & BEHAVIOR	SERVER & DATABASE EXECUTION
Pickup & Return Time Selector	Date and hour picker for vehicle collection and drop-off, with location pickup hub selection.	Calculates rental duration in 24-hour blocks; verifies vehicle availability against existing <code>RentalBooking</code> records.
Driving License Uploader	Front and back photo upload of valid Indian/International Driving License with OCR validation.	Scans binary magic-bytes, uploads to encrypted storage, and stages for host pre-handover verification.
Security Deposit Breakdown	Itemized breakdown showing Rental Base Rate, Refundable Security Deposit (₹3,000 - ₹10,000), and Insurance Add-on.	Creates separate ledger hold for security deposit to enable frictionless post-trip release upon clean inspection.

Screen 7: Traveler Wishlist & Saved Collections (/wishlist)

Personalization & Bookmarking Phase

Primary Function: Saved trip organizer allowing travelers to bookmark tours, activities, and rentals, organize them into folders, and track price changes.

UI COMPONENT / WIDGET	USER INTERACTION & BEHAVIOR	SERVER & DATABASE EXECUTION
Heart Icon Instant Toggle	One-click bookmarking on any listing card across catalog and detail views with micro-animation.	Executes optimistic UI update; dispatches <code>POST /api/wishlist</code> upserting into <code>WishlistItem</code> .
Categorized Folders	Custom folders (e.g. "Summer Trek 2027", "Goa Weekend") for grouping saved listings.	Relational foreign key query linking <code>WishlistItem</code> records by user and folder category.
Price Drop Alerts Badge	Visual badge indicating listings where host has discounted departure batch prices since bookmarking.	Compares listing <code>price</code> against <code>originalPrice</code> stored in database.

Screen 8: Trips & Orders Drawer / Management (/my-bookings)

Post-Booking Operations Phase

Primary Function: Central dashboard managing all confirmed, active, completed, and cancelled reservations across Tours, Activities, and Rentals.

UI COMPONENT / WIDGET	USER INTERACTION & BEHAVIOR	SERVER & DATABASE EXECUTION
Tabbed Status Filter	Tabs for "Upcoming", "Active / In-Progress", "Completed", and "Cancelled / Refunded".	Executes polymorphic queries across <code>TourBooking</code> , <code>ActivityBooking</code> , and <code>RentalBooking</code> sorted by date.
Digital Voucher & QR Code	Modal displaying booking confirmation voucher, QR code for host scan check-in, and PDF download button.	Generates cryptographically signed QR code containing <code>bookingCode</code> and encrypted traveler hash.
Cancellation & Refund Trigger	Tiered cancellation button displaying real-time refundable amount based on cancellation policy rules.	Executes cancellation policy calculation; initiates automated refund via <code>Razorpay.refunds</code> and logs to <code>Refund</code> .

Screen 9: Traveler Profile & Identity Settings (/profile)

Identity & Preference Governance

Primary Function: Traveler biographical data management, verified phone credentials, emergency contact synchronization, and DPDP Act consent history.

UI COMPONENT / WIDGET	USER INTERACTION & BEHAVIOR	SERVER & DATABASE EXECUTION
Avatar & Bio Editor	Photo upload with client-side cropping and short biographical description editor.	Scans image magic-bytes, optimizes via Cloudinary, and updates <code>UserProfile.avatarUrl</code> .
Phone OTP Verification	Input for 10-digit Indian mobile number with 6-digit OTP verification flow.	Generates cryptographic OTP, dispatches via SMS gateway, and sets <code>isPhoneVerified: true</code> upon confirmation.
DPDP Act Consent History Table	Read-only audit table showing exact dates, versions, and IP addresses when traveler consented to legal policies.	Queries <code>PolicyConsent</code> filtered by <code>userId</code> , displaying policy title, version, and consented timestamp.

Screen 10: Active Devices & Session Management (/settings/devices)

Account Security Governance

Primary Function: Multi-device session manager enforcing the 3-device maximum concurrent limit with remote revocation capabilities and FIFO eviction transparency.

UI COMPONENT / WIDGET	USER INTERACTION & BEHAVIOR	SERVER & DATABASE EXECUTION
Active Devices List	Cards for each active device showing device model, browser, IP address, approximate city, and "Current Device" badge.	Queries <code>UserDevice</code> joined with <code>Session</code> filtered by <code>userId</code> and <code>status = ACTIVE</code> (maximum 3 records).
Remote "Revoke Session" Button	Clicking terminates session on a remote device with immediate effect.	Revokes refresh token in Redis and sets <code>UserDevice.status = REVOKED</code> . Remote client receives HTTP 401 on next request.
FIFO Eviction Banner	Informational banner: "Travels Pro permits up to 3 concurrent devices. Logging in on a 4th device automatically evicts your oldest session."	Educational security UI minimizing user confusion when older tablets or secondary browsers are automatically logged out.

Screen 11: Multi-Dimensional Review & Rating Submission (/reviews/submit)

Reputation & Quality Loop

Primary Function: Post-trip feedback portal capturing verified reviews across 4 distinct quality dimensions with anti-review-bombing protection.

UI COMPONENT / WIDGET	USER INTERACTION & BEHAVIOR	SERVER & DATABASE EXECUTION
4-Dimension Rating Sliders	Independent 5-star ratings for: (1) Accuracy of Description, (2) Cleanliness & Safety, (3) Host Communication, (4) Value for Money.	Validates scores between 1.0 and 5.0; calculates weighted composite rating persisted to <code>Review</code> .
Trip Photo Uploader	Traveler attaches up to 6 authentic trip photos taken during the expedition.	Uploads to CDN with EXIF verification ensuring photos were captured during the actual booking timeframe.
Verified Traveler Gate	Form is accessible only if traveler has completed the booking (<code>status = COMPLETED</code>).	Backend verifies booking completion in PostgreSQL before accepting review submission; prevents fraudulent review generation.

Screen 12: Private Trip Circles Real-Time Chat (/tours/[slug]/chat)**Social Group Coordination Phase**

Primary Function: Private, end-to-end authenticated group chat room connecting confirmed tour travelers and their certified host via WebSockets.

UI COMPONENT / WIDGET	USER INTERACTION & BEHAVIOR	SERVER & DATABASE EXECUTION
Real-Time Chat Feed	Instant messaging feed with typing indicators, message read receipts, and member role badges ("Host", "Traveler").	Socket.IO client binds to <code>tour-\${tourId}</code> ; messages are broadcast to connected sockets and written to <code>TourMessage</code> .
Host Announcement Banner	Pinned banner at top of chat for host broadcast updates (e.g. "Departure time moved to 06:30 AM due to weather").	Restricted mutation: only users with <code>role = HOST</code> can publish pinned announcements. Dispatches push notifications.
Message Flagging & Reporting	Contextual menu on any message allowing travelers to report harassment or policy violations.	Creates <code>TourMessageReport</code> and automatically notifies admin moderation team for immediate review.

Screen 13: Dispute & Refund Mediation Modal (/disputes/new)**Resolution & Claims Phase**

Primary Function: Dispute initiation workbench allowing travelers to challenge host claims, report service non-delivery, or appeal security deposit withholdings.

UI COMPONENT / WIDGET	USER INTERACTION & BEHAVIOR	SERVER & DATABASE EXECUTION
Dispute Category Selector	Options: "Host No-Show", "Safety Violation", "Unfair Damage Claim", "Misrepresented Amenities", "Emergency Medical Cancellation".	Maps to <code>IncidentCategory</code> enum; attaches case priority based on severity.
Evidence Uploader	Multi-file uploader for GPS timestamps, photos, medical certificates, or police FIR reports.	Persists encrypted documents to secure evidence storage; logs to <code>IncidentEvent</code> .
Escrow Automatic Freeze	Notice confirming: "Host payout for this booking has been automatically frozen pending admin review."	Backend sets <code>Booking.isPayoutFrozen = true</code> , halting automated T+2 bank payout transfers until resolution.

Screen 14: Emergency Safety Center & SOS Broadcast (/safety/sos)**Emergency Operations Phase**

Primary Function: Critical safety interface providing one-tap emergency SOS broadcast, emergency contact dispatch, and direct link to local authorities.

UI COMPONENT / WIDGET	USER INTERACTION & BEHAVIOR	SERVER & DATABASE EXECUTION
One-Tap SOS Trigger Button	Prominent high-contrast red button with 3-second hold confirmation to prevent accidental activation.	Captures current high-accuracy GPS coordinates, dispatches automated SMS to emergency contacts, and creates PO <code>Incident</code> .
Emergency Hotline Directory	Direct dial buttons for Police (112), Ambulance (108), Tourist Helpline (1363), and Travels Pro 24x7 Safety Desk.	Native <code>tel:</code> protocol handler triggering immediate telephony dialer on mobile devices.
Live Location Share Link	Generates temporary, token-authenticated live tracking URL shareable via WhatsApp or SMS.	Streams real-time GPS telemetry to Redis geospatial key with 24-hour expiration TTL.

04 Host Persona Journey: 10 Screens, 6–Step KYC & Supply Studios

The host lifecycle provides rigorous identity vetting, statutory tax compliance, comprehensive listing authoring studios, two-stage vehicle custody inspections, and automated escrow payouts.

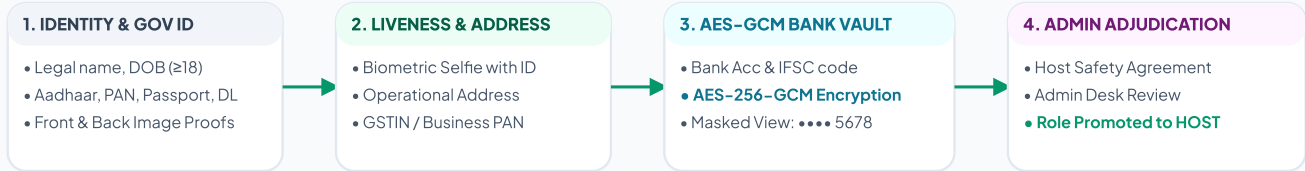


FIGURE 4.1: 6-STEP HOST ONBOARDING, BIOMETRIC VERIFICATION & ADJUDICATION PIPELINE

4.1 The 6–Step Host Identity & Banking Verification Wizard (/host/kyc)

To ensure traveler safety and prevent fraudulent operators, every host must complete a mandatory 6–stage verification dossier before their account can be promoted to `HOST` or publish listings.

KYC STEP	HOST INPUT REQUIREMENTS	VALIDATION & ENCRYPTION MECHANICS	DATABASE PERSISTENCE
Step 1: Legal Entity	Select <code>INDIVIDUAL</code> or <code>REGISTERED_BUSINESS</code> , Legal Full Name, Date of Birth, Business Trading Name.	Asserts age ≥ 18 years from DOB; ensures legal entity matches future tax filings and banking records.	<code>Host.hostType</code> , <code>Host.businessName</code>
Step 2: Tax Registration	Permanent Account Number (PAN) and optional Goods and Services Tax Identification Number (GSTIN).	Validates PAN format via regex (<code>^[A-Z]{5}[0-9]{4}[A-Z]{1}\$</code>); validates 15–character GSTIN structure.	<code>Host.panNumber</code> , <code>Host.gstNumber</code>
Step 3: Identity Document	Document Type (Aadhaar, Passport, Voter ID) and clear high-resolution front and back image uploads.	Scans binary magic-bytes (preventing disguised malware); stores images in encrypted private bucket.	<code>KycApplication.idType</code> , <code>idFrontUrl</code> , <code>idBackUrl</code>
Step 4: Liveness Selfie	Live webcam / smartphone capture of host holding their government ID card next to their face.	Performs facial symmetry and liveness check; verifies ID document visible in photo matches Step 3 upload.	<code>KycApplication.selfieUrl</code>
Step 5: Encrypted Bank	Bank Account Number, Confirm Account Number, Account Holder Name, Bank Name, and 11–character IFSC Code.	Validates IFSC format; encrypts account number in memory via AES-256-GCM (<code>\${iv}:\${ciphertext}:\${authTag}</code>).	<code>KycApplication.bankAccountEnc</code> , <code>bankIfsc</code>
Step 6: Statutory Clickwrap	Mandatory unchecked checkboxes for <code>HOST_SAFETY_AGREEMENT</code> and <code>TERMS_OF_SERVICE</code> .	Requires explicit affirmative consent; logs immutable IP, user-agent, and policy version record.	<code>PolicyConsent</code> (context: <code>HOST_KYC</code>)

4.2 Host Operational Studios & Management Screens (Screens 1 to 10)

Host Screen 1: Gateway & Fee Calculator (/host)

Onboarding landing page explaining platform reach, commission structure (12% Tours, 15% Activities, 18% Rentals), revenue calculator, and host type selector.

Host Screen 2: 6–Stage Identity & Banking KYC (/host/kyc)

Interactive multi-step wizard collecting legal identity, encrypted tax IDs, liveness selfie, and AES-256-GCM bank account credentials with progress indicator.

Host Screen 3: Host Mission Control Dashboard (/host/dashboard)

Real-time analytics dashboard displaying Gross Merchandise Value (GMV), 30-day net earnings, active guest roster, listing performance metrics, and pending booking alerts.

Host Screen 4: Tour Studio (7-Step Wizard) (/host/tours/create)

Comprehensive authoring studio: (1) General Info, (2) Day-by-Day Itinerary Builder, (3) Inclusions & Exclusions, (4) Departure Batches & Quotas, (5) Tiered Pricing, (6) Safety Gear & Disclosures, (7) Policy Confirmation. Listings are submitted to Admin Moderation Studio with `PENDING_REVIEW` status.

Host Screen 5: Hyperlocal Activity Slot Studio (/host/activities/manage)

Calendar management interface for single-day experiences allowing hosts to generate recurring daily time slots, set max capacity limits, configure instant booking toggles, and issue weather advisories.

Host Screen 6: Vehicle Fleet Inventory Manager (/host/rentals/fleet)

Automotive fleet inventory system tracking vehicle make, model, year, transmission, statutory Registration Certificate (RC), Pollution Under Control (PUC) certificate, and commercial insurance expiry dates.

Host Screen 7: Two-Stage Digital Custody Inspection Studio (/host/rentals/[id]/custody)

Mobile-optimized vehicle custody inspection tool. Eliminates security deposit disputes via two-stage photographic verification:

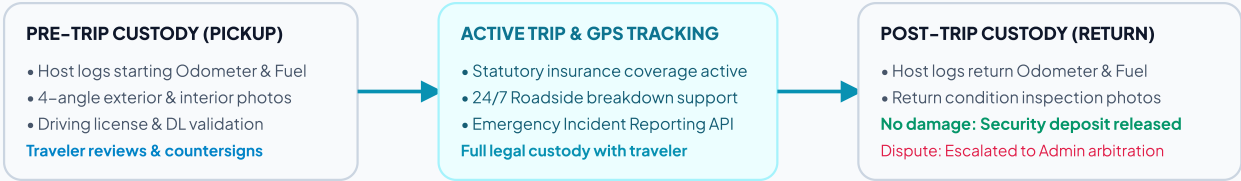


FIGURE 4.2: TWO-STAGE VEHICLE CUSTODY HANDOVER & INSPECTION ARCHITECTURE

INSPECTION STAGE	HOST OPERATIONAL CHECKLIST	TRAVELER VERIFICATION & LEGAL EFFECT
Stage 1: Pickup Handover	Host inputs starting odometer reading, fuel level (e.g. 100%), and captures 4 mandatory timestamped photos (Front, Rear, Left, Right).	Traveler physically inspects vehicle, reviews photos on mobile screen, and enters digital PIN/signature to accept custody. Transitions booking to <code>ACTIVE</code> .
Stage 2: Return Handover	Host inspects returned vehicle, logs return odometer, checks fuel level, and photographs vehicle condition.	If readings match terms and no damage is logged, security deposit is auto-released. If damage is detected, evidence is escalated to Admin Dispute Desk.

Host Screen 8: Booking Adjudication Queue (/host/bookings)

Real-time booking inbox displaying incoming reservation requests, guest verification status, special requests, and one-click Accept/Reject buttons with automated traveler SMS notifications.

Host Screen 9: Earnings & Automated Payout Ledger (/host/payouts)

Itemized financial ledger breaking down Gross Booking Value, Platform Commission (12-18%), GST Deductions (18%), and Net Settled Payouts. Integrates Razorpay Route for automated T+2 bank transfers upon tour completion.

Host Screen 10: Review Moderation & Dispute Console (/host/reviews)

Reputation management console allowing hosts to read traveler reviews, post professional public responses, and appeal fraudulent reviews to the admin moderation team.

05 Admin Persona Journey: 9 Screens, Governance & Safety Desk

The administrator persona enforces platform safety, regulatory compliance, content authenticity, financial integrity, and emergency incident dispatch. Privileged admin actions are protected by Step-Up Multi-Factor Authentication.

5.1 Exhaustive Catalog of Admin Mission Control Screens (Screens 1 to 9)

Admin Screen 1: Executive Mission Control (`/admin`)

High-level platform executive overview showing real-time Gross Merchandise Value (GMV), net platform commission revenue, active user counts, live trip circles, and a geographical safety incident heatmap.

Admin Screen 2: Step-Up Multi-Factor Authentication (`/admin/auth/step-up`)

Time-based One-Time Password (TOTP) / Biometric re-authentication barrier required before an administrator can approve host KYC, access decrypted bank numbers, or authorize financial refunds.

Admin Screen 3: Host KYC Adjudication Dossier (`/admin/kyc/[id]`)

Side-by-side inspection console: displays government ID uploads with high-resolution pan-and-zoom, live selfie comparison, automated PAN format verification, and decrypted AES-256-GCM bank details. Includes one-click "Approve & Promote to HOST" or "Reject with Stated Reason" buttons.

Admin Screen 4: Listing Moderation & Quality Studio (`/admin/listings`)

Review pipeline for newly submitted Tours, Activities, and Rentals. Administrators audit itinerary descriptions, verify photo authenticity against copyright infringements, detect price gouging, and toggle listing status (`APPROVED` , `REVISION_REQUESTED` , `ARCHIVED`).

Admin Screen 5: Legal Policy CMS & Versioning (`/admin/policies`)

Dynamic Markdown/WYSIWYG editor managing the 5 platform policies. When an admin increments a policy version (e.g. v1.0 → v1.1), the system automatically invalidates previous user consents and triggers the blocking `PolicyConsentModal` upon next login.

Admin Screen 6: Financial Reconciliation & Escrow Ledger (`/admin/financials`)

Comprehensive treasury dashboard tracking total funds held in Razorpay Escrow, pending T+2 host payout batches, automated GST liability calculations, and disputed transaction holds.

Admin Screen 7: Safety Incidents & Emergency SOS Desk (`/admin/incidents`)

Priority triage desk categorizing incidents by severity: P0 (Critical / Medical / Police), P1 (Vehicle Breakdown / Guide Misconduct), P2 (Booking Dispute). Displays live GPS telemetry, traveler emergency contacts, and incident timeline logger.

Admin Screen 8: Dispute & Damage Mediation Workbench (`/admin/disputes`)

Dual-pane evidence workbench displaying host pre/post-trip custody photos against traveler testimonies. Allows admins to authorize partial refunds, release security deposits, or order full traveler reimbursements.

Admin Screen 9: System Audit Log & B2B SSO Provisioning (`/admin/audit-ssu`)

Tamper-evident audit log streaming all administrative actions with actor ID, timestamp, IP address, and JSON diffs. Manages B2B corporate SSO configurations (SAML 2.0 / OIDC) and enterprise domain whitelists.

06 Core Subsystems Technical Deep Dive (What, When, How & Schemas)

This chapter provides an exhaustive technical specification of the **10 core operational subsystems** powering Travels Pro. Each module is documented across its architectural responsibility, trigger lifecycles, execution algorithms, request/response JSON schemas, state machines, concurrency controls, and failure recovery modes.

6.1 Module 1: Zero-Trust Authentication, Session & FIFO Device Eviction

DIMENSION	ENGINEERING SPECIFICATION
WHAT	Enterprise identity gateway managing credential authentication (bcrypt cost 12), NextAuth Google OAuth 2.0 handoffs, short-lived JWT access tokens, rotating refresh tokens, and strict 3-device maximum concurrent session governance.
WHEN	Triggered during user login (<code>/api/auth/login</code>), registration (<code>/api/auth/register</code>), Google OAuth callback (<code>/api/auth/google-login</code>), token refresh (<code>/api/auth/refresh</code>), and device management (<code>/api/auth/devices</code>).
HOW (Step-by-Step)	Credential Verification: Validates email format and asserts password ≥ 12 characters. Compares incoming password against <code>passwordHash</code> via constant-time bcrypt algorithm. FIFO Device Auto-Eviction: Queries <code>UserDevice</code> for existing active records. If active device count equals 3, the oldest device record (lowest <code>lastActiveAt</code>) is marked <code>status = EVICTED</code>, its corresponding refresh token is revoked in Redis/PostgreSQL, and the new device is inserted. Cryptographic Token Issuance: Generates a 15-minute HS256 JWT access token embedding <code>userId</code>, <code>role</code>, <code>email</code>, <code>sessionId</code>, and <code>deviceId</code>. Generates a cryptographically random 256-bit refresh token (7-day TTL) stored in secure, HttpOnly, SameSite=Lax cookie. Security Defense: Alg:none and asymmetric key confusion attacks are blocked by strict algorithm enforcement. Replay of revoked refresh tokens immediately invalidates the entire token family.
INPUT PAYLOAD (Request)	<pre>{ "email": "traveler@example.com", // string, email format "password": "Password123!@#", // string, min 12 chars "deviceId": "web-chrome-mac-893f", // string, unique client fingerprint "userAgent": "Mozilla/5.0 ...", // string, browser user-agent "ipAddress": "103.21.124.55" // string, IPv4/IPv6 client IP }</pre>
OUTPUT PAYLOAD (Response)	<pre>{ "success": true, "data": { "user": { "id": "usr_99f2a", "email": "traveler@example.com", "name": "Aarav Sharma", "role": "USER" }, "accessToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6Ikpz...", "expiresIn": 900, "activeDevices": 3, "evictedDeviceId": "web-safari-ios-112a" // null if under 3 devices } }</pre>
STATE MACHINE	<code>ACTIVE</code> → <code>EXPIRED</code> (TTL elapsed) <code>ACTIVE</code> → <code>EVICTED</code> (4th device login) <code>ACTIVE</code> → <code>REVOKED</code> (manual user logout or security breach).
CONCURRENCY CONTROLS	Concurrent logins across multiple tabs serialize on a row-level lock (<code>SELECT ... FOR UPDATE</code>) on <code>UserDevice</code> within a PostgreSQL transaction to prevent race-condition bypass of the 3-device limit.
FAILURE & FALLBACK	If Redis session cache is unreachable, authentication falls back gracefully to direct PostgreSQL database verification.

6.2 Module 2: Enterprise SSO & SAML 2.0 / OIDC Identity Federation

DIMENSION	ENGINEERING SPECIFICATION
WHAT	B2B identity federation subsystem enabling corporate enterprise accounts to authenticate employees through Okta, Microsoft Entra ID (Azure AD), Google Workspace, or Ping Identity.
WHEN	Initiated when an enterprise user accesses <code>/api/auth/sso/authorize</code> or returns via IdP callback to <code>/api/auth/sso/callback</code> .
HOW (Step-by-Step)	Domain Discovery: Resolves enterprise tenant from email domain (e.g. <code>@deloitte.com</code>) by querying <code>Organization.domain</code>. AuthnRequest Generation: Formulates Deflated, Base64-encoded XML SAML 2.0 AuthnRequest with HMAC-signed tamper-evident <code>RelayState</code>. Assertion Consumer Service (ACS): Parses signed SAML XML assertion, validates X.509 certificate signatures against stored IdP public keys, verifies <code>NotBefore</code> and <code>NotOnOrAfter</code> timestamps. Automated JIT Provisioning: Just-In-Time provisions user in PostgreSQL, assigns corporate travel policy roles, and maps enterprise cost-center metadata.
INPUT PAYLOAD (Request)	<pre>{ "SAMLResponse": "PHNhbWxwO1Jlc3BvbmlHhthb65z...", // Base64 encoded XML "RelayState": "hmac_signed_state_nonce_99182a" // Tamper-evident CSRF state }</pre>
OUTPUT PAYLOAD (Response)	<pre>{ "success": true, "data": { "user": { "id": "usr_sso_881", "email": "employee@corp.com", "role": "USER" }, "orgId": "org_deloitte_india", "sessionToken": "sso_sess_9912a441" } }</pre>
CONCURRENCY CONTROLS	JIT user provisioning utilizes PostgreSQL <code>ON CONFLICT (email) DO UPDATE</code> to eliminate duplicate user creation during parallel SSO callbacks.
FAILURE & FALLBACK	Expired SAML assertions return HTTP 401 <code>SSO_ASSERTION_EXPIRED</code> with an automated redirect back to the enterprise IdP sign-in gateway.

6.3 Module 3: Statutory DPDP Act 2023 Clickwrap & Legal Policy CMS

DIMENSION	ENGINEERING SPECIFICATION
WHAT	Regulatory legal consent engine enforcing explicit, voluntary clickwrap confirmations complying with India's Digital Personal Data Protection Act, 2023. Manages versioned policies and blocking re-consent prompts.
WHEN	Executes on user signup, host KYC onboarding, booking checkout, and upon administrative publishing of revised policy versions.
HOW (Step-by-Step)	Voluntary Clickwrap Enforcement: Pre-checked consent boxes are strictly prohibited by code. Checkboxes render unchecked. Immutable Audit Recording: When a user consents, <code>/api/policies/consent</code> writes an immutable record to <code>PolicyConsent</code> with fields: <code>userId</code>, <code>policyId</code>, <code>version</code>, <code>ipAddress</code>, <code>userAgent</code>, and <code>context</code> (<code>SIGNUP</code>, <code>HOST_KYC</code>, <code>BOOKING</code>). Dynamic Version Bumping: When an administrator edits policy text in <code>/admin/policies</code> and increments version (e.g. v1.0 → v1.1), subsequent login requests detect missing v1.1 consent, return <code>reconsentRequired: true</code>, and activate the blocking <code>PolicyConsentModal</code>.
INPUT PAYLOAD (Request)	<pre>{ "policyId": "pol_traveler_safety_v1", "version": "1.1", "context": "BOOKING", "consented": true }</pre>
OUTPUT PAYLOAD (Response)	<pre>{ "success": true, "data": { "consentId": "cns_991823a", "policyType": "TRAVELER_SAFETY_POLICY", "version": "1.1", "consentedAt": "2026-09-23T20:45:00.000Z" } }</pre>
CONCURRENCY CONTROLS	Compound database index on <code>[userId, policyId, version]</code> with unique constraint prevents duplicate consent records under rapid multi-clicks.
FAILURE & FALLBACK	Direct API mutations executed without active consent are intercepted by server middleware and aborted with HTTP 403 <code>POLICY_RECONSENT_REQUIRED</code> .

6.4 Module 4: 6-Stage Host KYC & Encrypted PII/Bank Vault

DIMENSION	ENGINEERING SPECIFICATION
WHAT	Host vetting engine verifying legal identity, date of birth, PAN/GSTIN, live biometric selfie holding ID, and AES-256-GCM encrypted bank account storage.
WHEN	Invoked during host application submission (<code>/api/host/kyc</code>) and administrative KYC adjudication (<code>/api/admin/kyc/[id]</code>).
HOW (Step-by-Step)	Statutory Input Sanitization: Verifies applicant is ≥ 18 years old. Validates PAN regex (<code>^[A-Z]{5}[0-9]{4}[A-Z]{1}\$</code>) and Aadhaar 12-digit format. Binary Magic-Byte File Scanning: Inspects uploaded image bytes to verify legitimate JPEG/PNG signatures, preventing executable code injection. AES-256-GCM Bank Encryption: Host bank account numbers are encrypted in memory using AES-256-GCM prior to database insertion. Stored as <code>\${iv}:\${ciphertext}:\${authTag}</code> . Database queries return only masked strings (<code>.... 5678</code>). Admin Adjudication: Admin adjudicators review credentials in a dual-pane console. Approving application automatically sets <code>Host.isVerified = true</code> and promotes user role from <code>USER</code> to <code>HOST</code> .
INPUT PAYLOAD (Request)	<pre>{ "hostType": "INDIVIDUAL", "panNumber": "ABCDE1234F", "idType": "AADHAAR", "idFrontUrl": "https://cdn.travelspro.com/kvc/front_9918.png", "selfieUrl": "https://cdn.travelspro.com/kyc/selfie_9918.jpg", "bankAccountNumber": "91823746192837", "bankIfsc": "HDFC0001234", "bankName": "HDFC Bank Ltd" }</pre>
OUTPUT PAYLOAD (Response)	<pre>{ "success": true, "data": { "kycId": "kyc_88192a", "status": "PENDING REVIEW", "bankAccountMasked": ".... 2837", "submittedAt": "2026-09-23T20:50:00.000Z" } }</pre>
STATE MACHINE	<code>DRAFT</code> → <code>PENDING_REVIEW</code> → <code>APPROVED</code> (Promoted to HOST) <code>REJECTED</code> (Reason provided, host resubmits) <code>SUSPENDED</code> .
CONCURRENCY CONTROLS	Host KYC table maintains a unique constraint on <code>userId</code> to ensure a user cannot have multiple simultaneous pending dossiers.
FAILURE & FALLBACK	Cryptographic verification tag failures throw structured <code>CryptoIntegrityError</code> ; plain-text credentials are never written to error logs.

6.5 Module 5: Multi-Day Group Tours & Departure Batch Scheduling Engine

DIMENSION	ENGINEERING SPECIFICATION
WHAT	Expedition scheduling system managing multi-day itineraries, discrete departure batches, seat capacity quotas, waitlist queues, and automated seat replenishment.
WHEN	Executes during tour authoring in Host Studio, catalog search, booking checkout, and cancellation workflows.
HOW (Step-by-Step)	Itinerary Modeling: Persists day-by-day activities to <code>TourItineraryDay</code> linked to parent <code>Tour</code>. Batch Scheduling: Hosts define departure batches (<code>TourBatch</code>) with independent <code>totalSlots</code> and <code>availableSlots</code>. Reservation Locking: Checkout creates an atomic <code>BookingIntent</code> lock, decrementing <code>availableSlots</code> for 15 minutes. Automated Waitlist Escalation: When a batch reaches 0 available slots, travelers can join <code>WaitlistQueue</code>. If a confirmed booking is cancelled, the first waitlisted user receives an exclusive 2-hour reservation window via SMS/Email.
INPUT PAYLOAD (Request)	<pre>{ "tourId": "tur_himalayan_trek_99", "batchId": "btc_oct_15_2026", "questCount": 2, "travelers": [{ "name": "Aarav Sharma", "age": 28, "gender": "MALE", "idNumber": "AADHAAR_9918" }, { "name": "Priya Patel", "age": 26, "gender": "FEMALE", "idNumber": "AADHAAR_9919" }], "emergencyContact": { "name": "Vikram Sharma", "phone": "9876543210" } }</pre>
OUTPUT PAYLOAD (Response)	<pre>{ "success": true, "data": { "bookingCode": "TRV-TOUR-8819", "status": "PENDING", "totalAmount": 24999.00, "currency": "INR", "lockExpiresAt": "2026-09-23T21:10:00.000Z" } }</pre>
STATE MACHINE	PENDING (15m intent) → CONFIRMED (payment success) → COMPLETED (tour ended) CANCELLED (refund processed).
CONCURRENCY CONTROLS	Slot decrements execute inside a PostgreSQL transaction using <code>UPDATE "TourBatch" SET "availableSlots" = "availableSlots" - \$1 WHERE id = \$2 AND "availableSlots" ≥ \$1</code> , preventing overbooking.
FAILURE & FALLBACK	If payment fails or expires, a scheduled BullMQ worker automatically replenishes available slots back to the batch.

6.6 Module 6: Hyperlocal Activities & Slot Scheduling Engine

DIMENSION	ENGINEERING SPECIFICATION
WHAT	Hyperlocal experience scheduling system managing recurring daily time slots (e.g. Scuba, Paragliding, Cooking Classes), participant quotas, and instant booking confirmations.
WHEN	Operates during host activity authoring, slot calendar generation, traveler checkout, and weather advisory alerts.
HOW (Step-by-Step)	Slot Recurrence Generator: Generates daily time-slot templates (<code>ActivitySlot</code>) with defined <code>startTime</code> , <code>endTime</code> , and <code>capacity</code> . Atomic Capacity Locking: Checkout decrements <code>availableSlots</code> atomically. Weather Emergency Halts: In case of adverse weather (e.g. high winds for paragliding), host can issue an emergency halt flag, triggering automatic 100% refund calculations for affected slot holders.
INPUT PAYLOAD (Request)	<pre>{ "activityId": "act_scuba_goa_11", "slotId": "slt_morning_0900", "questCount": 3, "date": "2026-10-10" }</pre>
OUTPUT PAYLOAD (Response)	<pre>{ "success": true, "data": { "bookingCode": "TRV-ACT-1192", "status": "CONFIRMED", "subtotal": 7500.00, "taxes": 1350.00, "totalAmount": 8850.00 } }</pre>
CONCURRENCY CONTROLS	Row-level lock on <code>ActivitySlot</code> enforces capacity boundaries without phantom reads.
FAILURE & FALLBACK	Host cancellations immediately credit the traveler's original payment method via Razorpay Refunds API without cancellation penalties.

6.7 Module 7: Car & Bike Rentals & Two-Stage Custody Handover

DIMENSION	ENGINEERING SPECIFICATION
WHAT	Fleet mobility operations engine enforcing statutory Registration Certificate (RC) validation, refundable security deposits, and a two-stage photographic custody handover system.
WHEN	Stage 1 executes at physical vehicle collection (Pickup); Stage 2 executes at physical vehicle return (Drop-off).
HOW (Step-by-Step)	Pre-Trip Handover (Stage 1): Host logs starting odometer reading and fuel level percentage, capturing 4 mandatory photos (Front, Rear, Left, Right). Traveler verifies on smartphone and enters digital PIN/signature to accept custody. Transitions booking to ACTIVE. Post-Trip Handover (Stage 2): Host logs return odometer, checks fuel level, and captures return condition photos. Deposit Settlement: If readings match agreed terms and no damage is logged, the security deposit is automatically refunded. If damage is reported, the case escalates to the Admin Dispute Desk with cryptographic timestamps.
INPUT PAYLOAD (Request)	<pre>{ "bookingId": "rnt_bk_99182", "stage": "PICKUP", "odometerReading": 45210, "fuelPercentage": 100, "photoUrls": ["https://cdn.travelspro.com/custody/front_9918.jpg", "https://cdn.travelspro.com/custody/rear_9918.jpg", "https://cdn.travelspro.com/custody/left_9918.jpg", "https://cdn.travelspro.com/custody/right_9918.jpg"], "travelerSignaturePin": "8921" }</pre>
OUTPUT PAYLOAD (Response)	<pre>{ "success": true, "data": { "inspectionId": "ins_pickup_8819", "bookingStatus": "ACTIVE", "custodyAcceptedAt": "2026-09-23T21:00:00.000Z" } }</pre>
STATE MACHINE	CONFIRMED → ACTIVE (Pickup inspection signed) → RETURN_INSPECTED → COMPLETED (Deposit released) DISPUTED (Damage claim).
CONCURRENCY CONTROLS	PostgreSQL unique constraint on [bookingId, stage] guarantees that an inspection stage cannot be recorded multiple times.
FAILURE & FALLBACK	Failure of host to record Stage 1 pickup photos legally bars host from claiming pre-existing scratches or damage upon return.

6.8 Module 8: Booking Intent, Inventory Lock & Razorpay Payment Escrow

DIMENSION	ENGINEERING SPECIFICATION
WHAT	Transaction orchestrator managing 15-minute temporary inventory locks, Razorpay payment order creation, HMAC-SHA256 signature verification, and automated escrow holdings.
WHEN	Active during checkout initiation, payment completion, and asynchronous Razorpay webhook delivery.
HOW (Step-by-Step)	Order Generation: Generates Razorpay Order ID via API with amount in paise (1 INR = 100 paise), attaching <code>bookingCode</code> as receipt. HMAC Signature Verification: Computes <code>crypto.createHmac('sha256', secret).update(order_id + ' ' + payment_id).digest('hex')</code> and asserts exact match against incoming client signature. Idempotent Webhook Processing: Webhook processor verifies <code>X-Razorpay-Signature</code>, logs payload to <code>PaymentWebhookEvent</code>, and transitions booking status to <code>CONFIRMED</code> within a database transaction.
INPUT PAYLOAD (Request)	<pre>{ "razorpay_order_id": "order_EKfMR192837", "razorpay_payment_id": "pay_29384710293", "razorpay_signature": "9a8b7c6d5e4f3a2b1c0d9e8f7a6b5c4d...", "idempotencyKey": "idem_88192a_99182" }</pre>
OUTPUT PAYLOAD (Response)	<pre>{ "success": true, "data": { "paymentId": "pay_internal_9918", "bookingStatus": "CONFIRMED", "amountPaid": 24999.00, "escrowStatus": "HELD" } }</pre>
CONCURRENCY CONTROLS	Unique index on <code>idempotencyKey</code> prevents double-capture or duplicate booking creation under flaky mobile network retries.
FAILURE & FALLBACK	Payment authorization failures immediately rollback the <code>BookingIntent</code> , restoring inventory to the public catalog.

6.9 Module 9: Trip Circles Real-Time WebSocket Communication

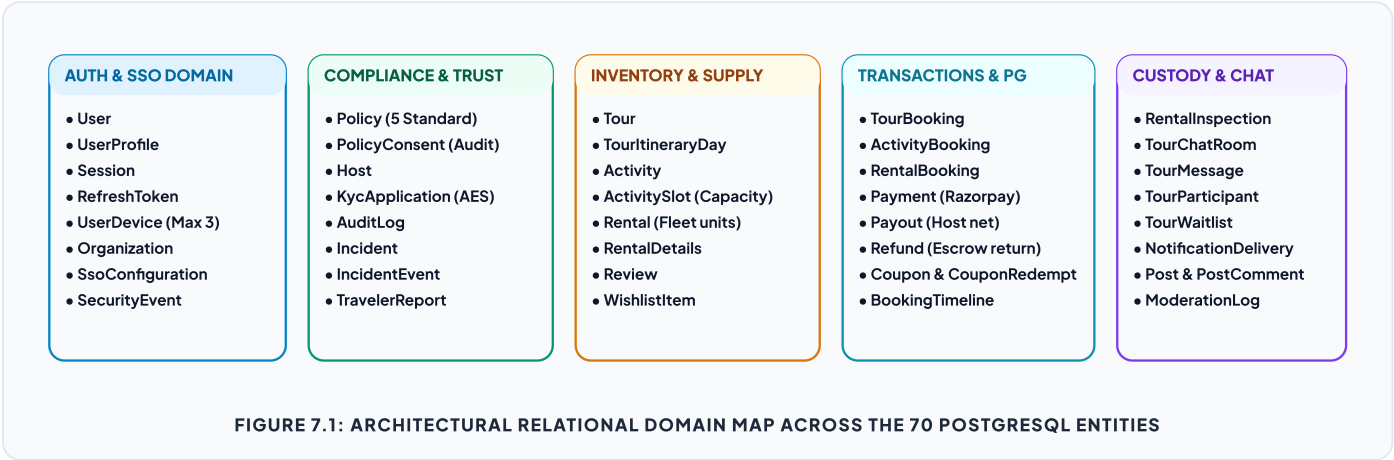
DIMENSION	ENGINEERING SPECIFICATION
WHAT	Private, authenticated WebSocket chat cluster allowing confirmed tour participants and their verified host to coordinate logistics, share itineraries, and broadcast announcements.
WHEN	Active continuously from booking confirmation through post-trip completion.
HOW (Step-by-Step)	Cookie-Based WebSocket Auth: Socket.IO handshake validates <code>travels_session</code> JWT cookie; rejects unauthorized connections. Dynamic Room Membership: Validates user has a confirmed booking for the target tour; joins socket to room <code>tour-\${tourId}</code>. Message Delivery & Persistence: Broadcasts messages to active room sockets and asynchronously writes to <code>TourMessage</code> with delivery timestamps. Moderation & Reporting: Travelers can report abusive messages, triggering automated admin notification and shadowban flags.
INPUT PAYLOAD (Socket Event: <code>send_message</code>)	<pre>{ "tourId": "tur_himalayan_trek_99", "content": "What is the recommended footwear for Day 3?", "type": "TEXT" // TEXT IMAGE ANNOUNCEMENT }</pre>
OUTPUT PAYLOAD (Socket Event: <code>new_message</code>)	<pre>{ "messageId": "msg_88192a", "tourId": "tur_himalayan_trek_99", "sender": { "id": "usr_99", "name": "Aarav Sharma", "role": "USER" }, "content": "What is the recommended footwear for Day 3?", "timestamp": "2026-09-23T21:05:00.000Z" }</pre>
CONCURRENCY CONTROLS	Redis Pub/Sub adapter coordinates WebSocket messaging across multi-instance serverless deployments without cross-server message drops.
FAILURE & FALLBACK	If WebSocket connection drops, client automatically falls back to long-polling and syncs missed messages via <code>/api/tour/[id]/chat</code> .

6.10 Module 10: Safety Incidents, SOS Telemetry & Governance

DIMENSION	ENGINEERING SPECIFICATION
WHAT	Emergency safety dispatch engine managing one-tap SOS alerts, emergency contact SMS broadcasts, GPS telemetry tracking, and administrative dispute mediation.
WHEN	Triggered when a traveler activates SOS (<code>/safety/sos</code>), reports an incident (<code>/api/incidents</code>), or disputes a rental damage charge.
HOW (Step-by-Step)	SOS Dispatch: Captures high-accuracy GPS coordinates, dispatches automated SMS via Twilio/Brevo to emergency contacts, and creates a PO Critical <code>Incident</code> in PostgreSQL. Admin Mission Control Alert: Real-time WebSocket triggers audio alarm and modal popup on active admin dashboards. Dispute Resolution Ledger: For financial disputes, admins inspect pre/post inspection photos, authorize partial or full refunds, and log immutable decisions to <code>AuditLog</code>.
INPUT PAYLOAD (Request: SOS Alert)	<pre>{ "bookingId": "tur_bk_88192", "latitude": 32.2396, "longitude": 77.1887, "accuracy": 4.5, "description": "Medical emergency on Rohtang pass trek" }</pre>
OUTPUT PAYLOAD (Response)	<pre>{ "success": true, "data": { "incidentId": "inc_00_88192a", "severity": "CRITICAL_P0", "dispatchedHotLines": ["112", "108"], "contactsNotifiedCount": 2, "status": "DISPATCHED" } }</pre>
STATE MACHINE	REPORTED → DISPATCHED → UNDER_INVESTIGATION → RESOLVED → CLOSED .
CONCURRENCY CONTROLS	Database write idempotency ensures that frantic multiple taps on the emergency SOS button update the existing incident location rather than generating duplicate dispatch tickets.
FAILURE & FALLBACK	If SMS gateway experiences latency, system automatically triggers automated phone call fallback to registered emergency contacts.

07 Exhaustive Database Data Dictionary (All 70 Models & 46 Enums)

The PostgreSQL database schema consists of **70 Prisma models** and **46 strict PostgreSQL enums** spanning 1,249 validated relational fields. The schema is normalized to 3NF/BCNF across seven operational domain clusters, incorporating compound B-Tree indexes, foreign key cascades, and column-level encryption.



Cryptographic & Storage Invariants:

- **AES-256-GCM Cryptographic Vault:** Fields suffixed with `Enc` (such as `bankAccountEnc` in `KycApplication` and SSO secrets in `SsoConfiguration`) are encrypted using AES-256-GCM with unique 96-bit initialization vectors and 128-bit authentication tags. Plaintext values are never written to disk or database logs.
- **Zero Mock / Strict Constraints:** All financial amounts are stored as `Decimal(10, 2)` to prevent floating-point rounding errors. All identity timestamps use UTC `DateTime`.
- **Idempotency Keys:** All commercial transaction tables (`TourBooking`, `ActivityBooking`, `RentalBooking`) enforce a unique `idempotencyKey` constraint to guarantee zero double-charging under network retries.

7.1 Domain 4: Hyperlocal Activities & Slot Scheduling (7 Relational Models)

This domain encompasses 7 relational entities enforcing data integrity, foreign key cascades, and business constraints.

Model: Activity

52 Attributes • 5 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
hostId	String	—	Core entity attribute
slug	String	@unique	Unique constraint
title	String	—	Core entity attribute
description	String?	—	Core entity attribute
city	String	—	Core entity attribute
state	String?	—	Core entity attribute
country	String	@default("India")	Core entity attribute
area	String?	—	Core entity attribute
category	ActivityCategory	—	Core entity attribute
difficulty	ActivityDifficulty	@default(EASY)	Core entity attribute
price	Decimal	@db.Decimal(10, 2)	Monetary currency value (INR)
originalPrice	Decimal?	@db.Decimal(10, 2)	Core entity attribute
duration	String	—	Core entity attribute
groupSizeMin	Int	@default(1)	Core entity attribute
groupSizeMax	Int	@default(20)	Core entity attribute
totalSlots	Int	@default(20)	Core entity attribute
availableSlots	Int	@default(20)	Core entity attribute
language	String	@default("English")	Core entity attribute
imageUrl	String?	—	Core entity attribute
images	String[]	—	Core entity attribute
highlights	String[]	—	Core entity attribute
included	String[]	—	Core entity attribute
excluded	String[]	—	Core entity attribute
meetingPoint	String?	—	Core entity attribute
meetingLat	Float?	—	Core entity attribute
meetingLng	Float?	—	Core entity attribute
cancellationPolicy	String?	—	Core entity attribute
status	ListingStatus	@default(PENDING_REVIEW)	State machine lifecycle enum
isActive	Boolean	@default(true)	Boolean state toggle
isApproved	Boolean	@default(false)	Boolean state toggle
averageRating	Float	@default(0)	Core entity attribute
totalReviews	Int	@default(0)	Core entity attribute
totalBookings	Int	@default(0)	Core entity attribute
submittedForReviewAt	DateTime?	—	Timestamp lifecycle marker
approvedAt	DateTime?	—	Timestamp lifecycle marker
rejectedAt	DateTime?	—	Timestamp lifecycle marker
archivedAt	DateTime?	—	Timestamp lifecycle marker

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
<code>deletedAt</code>	<code>DateTime?</code>	—	Timestamp lifecycle marker
<code>reviewedAt</code>	<code>DateTime?</code>	—	Timestamp lifecycle marker
<code>reviewedById</code>	<code>String?</code>	—	Core entity attribute
<code>rejectionReason</code>	<code>String?</code>	—	Core entity attribute
<code>moderationNotes</code>	<code>String?</code>	—	Core entity attribute
<code>lastReviewedStatus</code>	<code>ListingStatus?</code>	—	State machine lifecycle enum
<code>createdAt</code>	<code>DateTime</code>	@default(now())	Timestamp lifecycle marker
<code>updatedAt</code>	<code>DateTime</code>	@updatedAt	Timestamp lifecycle marker
<code>Host</code>	<code>Host</code>	@relation(fields: [hostId], references: [id], onDelete: Cascade)	Foreign key relationship
<code>ReviewedBy</code>	<code>User?</code>	@relation("ActivityReviewer", fields: [reviewedById], references: [id])	Foreign key relationship
<code>ActivityBooking</code>	<code>ActivityBooking[]</code>	—	Core entity attribute
<code>ActivitySlot</code>	<code>ActivitySlot[]</code>	—	Core entity attribute
<code>Review</code>	<code>Review[]</code>	—	Core entity attribute
<code>WishlistItem</code>	<code>WishlistItem[]</code>	—	Core entity attribute

Indexes & Constraints: @@index([city, category]) | @@index([hostId]) | @@index([status, isActive, isApproved]) | @@index([reviewedById]) | @@index([deletedAt])

Model: ActivityBooking

34 Attributes • 5 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
bookingCode	String	@unique	Unique constraint
idempotencyKey	String?	@unique	Unique constraint
userId	String	—	Core entity attribute
hostId	String	—	Core entity attribute
activityId	String	—	Core entity attribute
slotId	String?	—	Core entity attribute
date	DateTime	—	Core entity attribute
startTime	String	—	Core entity attribute
guestCount	Int	@default(1)	Core entity attribute
privateGroup	Boolean	@default(false)	Core entity attribute
subtotal	Decimal	@db.Decimal(10, 2)	Core entity attribute
taxes	Decimal	@default(0) @db.Decimal(10, 2)	Core entity attribute
totalAmount	Decimal	@db.Decimal(10, 2)	Monetary currency value (INR)
currency	String	@default("INR")	Core entity attribute
status	BookingStatus	@default(PENDING)	State machine lifecycle enum
expiresAt	DateTime?	—	Timestamp lifecycle marker
contactName	String	—	Core entity attribute
contactEmail	String	—	Core entity attribute
contactPhone	String	—	Core entity attribute
specialRequests	String?	—	Core entity attribute
cancellationReason	String?	—	Core entity attribute
cancelledAt	DateTime?	—	Timestamp lifecycle marker
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
Activity	Activity	@relation(fields: [activityId], references: [id])	Foreign key relationship
Host	Host	@relation(fields: [hostId], references: [id])	Foreign key relationship
ActivitySlot	ActivitySlot?	@relation(fields: [slotId], references: [id])	Foreign key relationship
User	User	@relation(fields: [userId], references: [id])	Foreign key relationship
ActivityBookingGuest	ActivityBookingGuest[]	—	Core entity attribute
ActivityBookingTimeline	ActivityBookingTimeline[]	—	Core entity attribute
ActivityPayment	ActivityPayment?	—	Core entity attribute
ActivityRefund	ActivityRefund?	—	Core entity attribute
Review	Review?	—	Core entity attribute

Indexes & Constraints: @@index([activityId]) | @@index([hostId]) | @@index([status]) | @@index([userId]) | @@index([expiresAt])

Model: ActivityBookingGuest

8 Attributes • 1 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
bookingId	String	—	Core entity attribute
type	GuestType	@default(ADULT)	Core entity attribute
firstName	String	—	Core entity attribute
lastName	String?	—	Core entity attribute
age	Int?	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
ActivityBooking	ActivityBooking	@relation(fields: [bookingId], references: [id], onDelete: Cascade)	Foreign key relationship

Indexes & Constraints: @@index([bookingId])

Model: ActivityBookingTimeline

8 Attributes • 1 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
bookingId	String	—	Core entity attribute
type	BookingEventType	—	Core entity attribute
title	String	—	Core entity attribute
message	String?	—	Core entity attribute
metadata	Json?	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
ActivityBooking	ActivityBooking	@relation(fields: [bookingId], references: [id], onDelete: Cascade)	Foreign key relationship

Indexes & Constraints: @@index([bookingId])

Model: ActivityPayment

15 Attributes • 1 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
activityBookingId	String	@unique	Unique constraint
userId	String	—	Core entity attribute
amount	Decimal	@db.Decimal(10, 2)	Core entity attribute
currency	String	@default("INR")	Core entity attribute
status	PaymentStatus	@default(PENDING)	State machine lifecycle enum
provider	String	@default("razorpay")	Core entity attribute
providerOrderId	String?	@unique	Unique constraint
providerPaymentId	String?	@unique	Unique constraint
transactionId	String?	@unique	Unique constraint
paidAt	DateTime?	—	Timestamp lifecycle marker
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
ActivityBooking	ActivityBooking	@relation(fields: [activityBookingId], references: [id], onDelete: Cascade)	Foreign key relationship
ActivityRefund	ActivityRefund?	—	Core entity attribute

Indexes & Constraints: @@index([status])

Model: ActivityRefund

15 Attributes • 1 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
activityBookingId	String	@unique	Unique constraint
activityPaymentId	String?	@unique	Unique constraint
requestedAmount	Decimal	@db.Decimal(10, 2)	Monetary currency value (INR)
approvedAmount	Decimal?	@db.Decimal(10, 2)	Monetary currency value (INR)
currency	String	@default("INR")	Core entity attribute
status	String	@default("REVIEW_PENDING")	State machine lifecycle enum
reason	String?	—	Core entity attribute
policySnapshot	String	—	Core entity attribute
providerRefundId	String?	—	Core entity attribute
failureReason	String?	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
processedAt	DateTime?	—	Timestamp lifecycle marker
ActivityBooking	ActivityBooking	@relation(fields: [activityBookingId], references: [id], onDelete: Cascade)	Foreign key relationship
ActivityPayment	ActivityPayment?	@relation(fields: [activityPaymentId], references: [id])	Foreign key relationship

Indexes & Constraints: @@index([status, createdAt])

Model: ActivitySlot

11 Attributes • 2 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
activityId	String	—	Core entity attribute
date	DateTime	@db.Date	Core entity attribute
startTime	String	—	Core entity attribute
totalSpots	Int	@default(20)	Core entity attribute
bookedSpots	Int	@default(0)	Core entity attribute
isActive	Boolean	@default(true)	Boolean state toggle
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
ActivityBooking	ActivityBooking[]	—	Core entity attribute
Activity	Activity	@relation(fields: [activityId], references: [id], onDelete: Cascade)	Foreign key relationship
Indexes & Constraints: @@unique([activityId, date, startTime]) @@index([activityId, date])			

7.2 Domain 1: Identity, Auth & Enterprise SSO (17 Relational Models)

This domain encompasses 17 relational entities enforcing data integrity, foreign key cascades, and business constraints.

Model: AdminProfile

7 Attributes • 0 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
userId	String	@unique	Unique constraint
department	String?	—	Core entity attribute
permissions	Json?	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
User	User	@relation(fields: [userId], references: [id], onDelete: Cascade)	Foreign key relationship

Model: RefreshToken

12 Attributes • 4 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
userId	String	—	Core entity attribute
tokenHash	String	@unique	Unique constraint
deviceId	String?	—	Core entity attribute
deviceName	String?	—	Core entity attribute
ipAddress	String?	—	Core entity attribute
userAgent	String?	—	Core entity attribute
isRevoked	Boolean	@default(false)	Boolean state toggle
revokedAt	DateTime?	—	Timestamp lifecycle marker
expiresAt	DateTime	—	Timestamp lifecycle marker
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
User	User	@relation(fields: [userId], references: [id], onDelete: Cascade)	Foreign key relationship

Indexes & Constraints: @@index([userId]) | @@index([tokenHash]) | @@index([expiresAt]) | @@index([isRevoked])

Model: Session

11 Attributes • 4 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
userId	String	—	Core entity attribute
deviceId	String?	—	Core entity attribute
deviceName	String?	—	Core entity attribute
deviceType	String?	—	Core entity attribute
ipAddress	String?	—	Core entity attribute
userAgent	String?	—	Core entity attribute
lastSeenAt	DateTime	@default(now())	Timestamp lifecycle marker
isActive	Boolean	@default(true)	Boolean state toggle
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
User	User	@relation(fields: [userId], references: [id], onDelete: Cascade)	Foreign key relationship

Indexes & Constraints: @@index([userId]) | @@index([deviceId]) | @@index([isActive]) | @@index([lastSeenAt])

Model: OtpVerification

9 Attributes • 4 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
email	String?	—	Core entity attribute
phone	String?	—	Core entity attribute
otpHash	String	—	Core entity attribute
purpose	OtpPurpose	—	Core entity attribute
attempts	Int	@default(0)	Core entity attribute
expiresAt	DateTime	—	Timestamp lifecycle marker
verifiedAt	DateTime?	—	Timestamp lifecycle marker
createdAt	DateTime	@default(now())	Timestamp lifecycle marker

Indexes & Constraints: @@index([email]) | @@index([phone]) | @@index([purpose]) | @@index([expiresAt])

Model: PasswordResetToken

7 Attributes • 4 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
userId	String	—	Core entity attribute
tokenHash	String	@unique	Unique constraint
expiresAt	DateTime	—	Timestamp lifecycle marker
usedAt	DateTime?	—	Timestamp lifecycle marker
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
User	User	@relation(fields: [userId], references: [id], onDelete: Cascade)	Foreign key relationship

Indexes & Constraints: @@index([userId]) | @@index([tokenHash]) | @@index([expiresAt]) | @@index([usedAt])

Model: LoginAttempt

5 Attributes • 4 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
email	String?	—	Core entity attribute
ipAddress	String	—	Core entity attribute
success	Boolean	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
Indexes & Constraints: @@index([email]) @@index([ipAddress]) @@index([success]) @@index([createdAt])			

Model: UserDevice

15 Attributes • 4 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
userId	String	—	Core entity attribute
deviceId	String	—	Core entity attribute
deviceName	String?	—	Core entity attribute
deviceType	String	@default("browser")	Core entity attribute
browser	String?	—	Core entity attribute
os	String?	—	Core entity attribute
ipAddress	String?	—	Core entity attribute
userAgent	String?	—	Core entity attribute
lastSeenAt	DateTime	@default(now())	Timestamp lifecycle marker
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
isCurrent	Boolean	@default(false)	Boolean state toggle
isActive	Boolean	@default(true)	Boolean state toggle
refreshTokenHash	String?	—	Core entity attribute
User	User	@relation(fields: [userId], references: [id], onDelete: Cascade)	Foreign key relationship
Indexes & Constraints: @@unique([userId, deviceId]) @@index([userId]) @@index([deviceId]) @@index([isActive])			

Model: EmailChangeRequest

8 Attributes • 3 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
userId	String	—	Core entity attribute
oldEmail	String	—	Core entity attribute
newEmail	String	—	Core entity attribute
otpHash	String	—	Core entity attribute
expiresAt	DateTime	—	Timestamp lifecycle marker
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
User	User	@relation(fields: [userId], references: [id], onDelete: Cascade)	Foreign key relationship

Indexes & Constraints: @@index([userId]) | @@index([newEmail]) | @@index([expiresAt])

Model: SecurityEvent

8 Attributes • 3 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
userId	String?	—	Core entity attribute
type	String	—	Core entity attribute
metadata	Json?	—	Core entity attribute
ipAddress	String?	—	Core entity attribute
userAgent	String?	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
User	User?	@relation(fields: [userId], references: [id], onDelete: SetNull)	Foreign key relationship

Indexes & Constraints: @@index([userId]) | @@index([type]) | @@index([createdAt])

Model: UserBlock

6 Attributes • 2 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
blockerId	String	—	Core entity attribute
blockedId	String	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
Blocker	User	@relation("UserBlocksCreated", fields: [blockerId], references: [id], onDelete: Cascade)	Foreign key relationship
Blocked	User	@relation("UserBlocksReceived", fields: [blockedId], references: [id], onDelete: Cascade)	Foreign key relationship

Indexes & Constraints: @@unique([blockerId, blockedId]) | @@index([blockedId])

Model: User

81 Attributes • 8 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
name	String	—	Core entity attribute
email	String	@unique	Unique constraint
password	String?	—	Core entity attribute
phone	String?	@unique	Unique constraint
role	Role	@default(USER)	Core entity attribute
status	AccountStatus	@default(ACTIVE)	State machine lifecycle enum
isEmailVerified	Boolean	@default(false)	Boolean state toggle
emailVerifiedAt	DateTime?	—	Timestamp lifecycle marker
isPhoneVerified	Boolean	@default(false)	Boolean state toggle
phoneVerifiedAt	DateTime?	—	Timestamp lifecycle marker
failedLoginAttempts	Int	@default(0)	Core entity attribute
lockedUntil	DateTime?	—	Core entity attribute
verificationToken	String?	—	Core entity attribute
verificationExpiry	DateTime?	—	Core entity attribute
resetToken	String?	—	Core entity attribute
resetTokenExpiry	DateTime?	—	Core entity attribute
provider	String	@default("credentials")	Core entity attribute
providerId	String?	—	Core entity attribute
lastLoginAt	DateTime?	—	Timestamp lifecycle marker
sessionInvalidatedAt	DateTime?	—	Timestamp lifecycle marker
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
isActive	Boolean	@default(true)	Boolean state toggle
isBanned	Boolean	@default(false)	Boolean state toggle
moderationReason	String?	—	Core entity attribute
reviewedAt	DateTime?	—	Timestamp lifecycle marker
reviewedById	String?	—	Core entity attribute
bannedAt	DateTime?	—	Timestamp lifecycle marker
suspendedAt	DateTime?	—	Timestamp lifecycle marker
deletedAt	DateTime?	—	Timestamp lifecycle marker
ActivityBooking	ActivityBooking[]	—	Core entity attribute
AdminProfile	AdminProfile?	—	Core entity attribute
AuditLog	AuditLog[]	—	Core entity attribute
Booking	Booking[]	—	Core entity attribute
TourParticipant	TourParticipant[]	—	Core entity attribute
SentMessages	TourMessage[]	—	Core entity attribute
SeenTourMessages	TourMessageSeen[]	—	Core entity attribute

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
TravelPreferences	UserTravelPreference?	—	Core entity attribute
ReportedTravelers	TravelerReport[]	@relation("Reporter")	Foreign key relationship
ReceivedReports	TravelerReport[]	@relation("ReportedUser")	Foreign key relationship
BlocksCreated	UserBlock[]	@relation("UserBlocksCreated")	Foreign key relationship
BlocksReceived	UserBlock[]	@relation("UserBlocksReceived")	Foreign key relationship
MessageReportsMade	TourMessageReport[]	@relation("TourMessageReportReporter")	Foreign key relationship
MessageReportsAbout	TourMessageReport[]	@relation("TourMessageReportSubject")	Foreign key relationship
MessageReportsReviewed	TourMessageReport[]	@relation("TourMessageReportReviewer")	Foreign key relationship
CouponRedemption	CouponRedemption[]	—	Core entity attribute
Host	Host?	—	Core entity attribute
Notification	Notification[]	—	Core entity attribute
NotificationDelivery	NotificationDelivery[]	—	Core entity attribute
Payment	Payment[]	—	Core entity attribute
Payout	Payout[]	—	Core entity attribute
Post	Post[]	—	Core entity attribute
PostComment	PostComment[]	—	Core entity attribute
PostLike	PostLike[]	—	Core entity attribute
RefreshToken	RefreshToken[]	—	Core entity attribute
Session	Session[]	—	Core entity attribute
PasswordResetToken	PasswordResetToken[]	—	Core entity attribute
UserDevice	UserDevice[]	—	Core entity attribute
EmailChangeRequest	EmailChangeRequest[]	—	Core entity attribute
SecurityEvent	SecurityEvent[]	—	Core entity attribute
RentalBooking	RentalBooking[]	—	Core entity attribute
Review	Review[]	—	Core entity attribute
UserProfile	UserProfile?	—	Core entity attribute
WishlistItem	WishlistItem[]	—	Core entity attribute
TourWaitlist	TourWaitlist[]	—	Core entity attribute
ReviewedUsers	User[]	@relation("UserReviewer")	Foreign key relationship
ReviewedByUser	User?	@relation("UserReviewer", fields: [reviewedById], references: [id])	Foreign key relationship
ReviewedHosts	Host[]	@relation("HostReviewer")	Foreign key relationship
ReviewedTours	Tour[]	@relation("TourReviewer")	Foreign key relationship
ReviewedRentals	Rental[]	@relation("RentalReviewer")	Foreign key relationship
ReviewedActivities	Activity[]	@relation("ActivityReviewer")	Foreign key relationship
ReviewedKyc	KycApplication[]	@relation("KycReviewer")	Foreign key relationship
ModerationLogs	ModerationLog[]	@relation("ModerationLogAdmin")	Foreign key relationship
ReportedIncidents	Incident[]	@relation("IncidentReporter")	Foreign key relationship
OwnedIncidents	Incident[]	@relation("IncidentOwner")	Foreign key relationship

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
IncidentEvents	IncidentEvent[]	@relation("IncidentActor")	Foreign key relationship
OrganizationMember	OrganizationMember[]	—	Core entity attribute
SsoAuditLog	SsoAuditLog[]	—	Core entity attribute
PolicyConsents	PolicyConsent[]	—	Core entity attribute
PublishedPolicies	Policy[]	@relation("PolicyPublisher")	Foreign key relationship

Indexes & Constraints: @@index([email]) | @@index([role]) | @@index([status]) | @@index([lockedUntil]) | @@index([isBanned]) | @@index([isActive]) | @@index([deletedAt]) | @@index([reviewedById])

Model: **UserProfile**

36 Attributes • 0 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
userId	String	@unique	Unique constraint
avatarUrl	String?	—	Core entity attribute
bio	String?	—	Core entity attribute
dateOfBirth	DateTime?	—	Core entity attribute
gender	Gender?	—	Core entity attribute
address	String?	—	Core entity attribute
city	String?	—	Core entity attribute
state	String?	—	Core entity attribute
country	String?	—	Core entity attribute
postalCode	String?	—	Core entity attribute
themeSettings	Json?	@map("theme")	Core entity attribute
notificationSettings	Json?	@map("notification")	Core entity attribute
uiSettings	Json?	@map("ui")	Core entity attribute
preferences	Json?	—	Core entity attribute
showGender	Boolean	@default(true)	Core entity attribute
showCity	Boolean	@default(true)	Core entity attribute
showInstagram	Boolean	@default(false)	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
User	User	@relation(fields: [userId], references: [id], onDelete: Cascade)	Foreign key relationship
languages	String[]	—	Core entity attribute
interests	String[]	—	Core entity attribute
travelStyle	TravelStyle?	—	Core entity attribute
personalityType	String?	—	Core entity attribute
foodPreference	String?	—	Core entity attribute
smokingPreference	Boolean?	—	Core entity attribute
drinkingPreference	Boolean?	—	Core entity attribute
sleepSchedule	String?	—	Core entity attribute
instagramHandle	String?	—	Core entity attribute
travelExperience	String?	—	Core entity attribute
trustScore	Int	@default(0)	Core entity attribute
emergencyContactName	String?	—	Core entity attribute
emergencyContactPhone	String?	—	Core entity attribute
soloTraveler	Boolean	@default(false)	Core entity attribute
womenSafetyVerified	Boolean	@default(false)	Core entity attribute

Model: UserTravelPreference

11 Attributes • 0 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
userId	String	@unique	Unique constraint
preferredAgeMin	Int?	—	Core entity attribute
preferredAgeMax	Int?	—	Core entity attribute
preferredLanguages	String[]	—	Core entity attribute
interests	String[]	—	Core entity attribute
likesAdventure	Boolean	@default(false)	Core entity attribute
likesNightlife	Boolean	@default(false)	Core entity attribute
likesNature	Boolean	@default(false)	Core entity attribute
budgetRange	String?	—	Core entity attribute
User	User	@relation(fields: [userId], references: [id])	Foreign key relationship

Model: Organization

10 Attributes • 1 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
slug	String	@unique	Unique constraint
name	String	—	Core entity attribute
domain	String	@unique	Unique constraint
isActive	Boolean	@default(true)	Boolean state toggle
ssoConfig	SsoConfiguration?	—	Core entity attribute
members	OrganizationMember[]	—	Core entity attribute
auditLogs	SsoAuditLog[]	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker

Indexes & Constraints: @@index([domain, isActive])

Model: SsoConfiguration

19 Attributes • 1 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
organizationId	String	@unique	Unique constraint
organization	Organization	@relation(fields: [organizationId], references: [id], onDelete: Cascade)	Foreign key relationship
protocol	SsoProtocol	@default(OIDC)	Core entity attribute
provider	SsoIdpProvider	@default(ENTRA_ID)	Core entity attribute
issuerUrl	String?	—	Boolean state toggle
clientId	String?	—	Core entity attribute
clientSecretEncrypted	String?	—	AES-256-GCM encrypted field
entryPoint	String?	—	Core entity attribute
issuer	String?	—	Boolean state toggle
cert	String?	@db.Text	Core entity attribute
logoutUrl	String?	—	Core entity attribute
jitEnabled	Boolean	@default(true)	Core entity attribute
defaultRole	Role	@default(USER)	Core entity attribute
inactivityTimeoutMinutes	Int	@default(30)	Core entity attribute
enforceMfaAtIdp	Boolean	@default(true)	Core entity attribute
allowPasswordFallback	Boolean	@default(false)	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
Indexes & Constraints: @@index([organizationId])			

Model: OrganizationMember

11 Attributes • 4 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
organizationId	String	—	Core entity attribute
userId	String	—	Core entity attribute
employeeId	String?	—	Core entity attribute
department	String?	—	Core entity attribute
role	String	@default("EMPLOYEE")	Core entity attribute
isCurrent	Boolean	@default(true)	Boolean state toggle
joinedAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
organization	Organization	@relation(fields: [organizationId], references: [id], onDelete: Cascade)	Foreign key relationship
user	User	@relation(fields: [userId], references: [id], onDelete: Cascade)	Foreign key relationship
Indexes & Constraints: @@unique([organizationId, userId]) @@unique([organizationId, employeeId]) @@index([organizationId, isCurrent]) @@index([userId])			

Model: SsoAuditLog

14 Attributes • 3 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
organizationId	String	—	Core entity attribute
userId	String?	—	Core entity attribute
email	String	—	Core entity attribute
eventType	String	—	Core entity attribute
protocol	SsoProtocol	—	Core entity attribute
idpEntityId	String?	—	Core entity attribute
ipAddress	String?	—	Core entity attribute
userAgent	String?	—	Core entity attribute
failureReason	String?	—	Core entity attribute
metadata	Json?	—	Core entity attribute
timestamp	DateTime	@default(now())	Core entity attribute
organization	Organization	@relation(fields: [organizationId], references: [id], onDelete: Cascade)	Foreign key relationship
user	User?	@relation(fields: [userId], references: [id], onDelete: SetNull)	Foreign key relationship

Indexes & Constraints: @@index([organizationId, timestamp]) | @@index([email, eventType]) | @@index([userId])

7.3 Domain 6: Financial Ledgers, Escrow & Coupons (6 Relational Models)

This domain encompasses 6 relational entities enforcing data integrity, foreign key cascades, and business constraints.

Model: Amenity

4 Attributes • 0 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
name	String	@unique	Unique constraint
icon	String?	—	Core entity attribute
category	String?	—	Core entity attribute

Model: Coupon

18 Attributes • 1 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
code	String	@unique	Unique constraint
name	String	—	Core entity attribute
description	String?	—	Core entity attribute
type	CouponType	—	Core entity attribute
value	Decimal	@db.Decimal(10, 2)	Core entity attribute
maxDiscount	Decimal?	@db.Decimal(10, 2)	Core entity attribute
minOrderAmount	Decimal	@default(0) @db.Decimal(10, 2)	Monetary currency value (INR)
usageLimit	Int?	—	Core entity attribute
usedCount	Int	@default(0)	Core entity attribute
perUserLimit	Int	@default(1)	Core entity attribute
startsAt	DateTime?	—	Timestamp lifecycle marker
expiresAt	DateTime?	—	Timestamp lifecycle marker
isActive	Boolean	@default(true)	Boolean state toggle
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
Booking	Booking[]	—	Core entity attribute
CouponRedemption	CouponRedemption[]	—	Core entity attribute

Indexes & Constraints: @@index([isActive, expiresAt])

Model: CouponRedemption

9 Attributes • 3 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
couponId	String	—	Core entity attribute
userId	String	—	Core entity attribute
bookingId	String?	—	Core entity attribute
amount	Decimal	@db.Decimal(10, 2)	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
Booking	Booking?	@relation(fields: [bookingId], references: [id])	Foreign key relationship
Coupon	Coupon	@relation(fields: [couponId], references: [id])	Foreign key relationship
User	User	@relation(fields: [userId], references: [id])	Foreign key relationship
Indexes & Constraints: @@unique([couponId, userId, bookingId]) @@index([couponId]) @@index([userId])			

Model: Payment

21 Attributes • 3 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
bookingId	String	@unique	Unique constraint
userId	String	—	Core entity attribute
hostId	String	—	Core entity attribute
amount	Decimal	@db.Decimal(10, 2)	Core entity attribute
hostEarnings	Decimal	@db.Decimal(10, 2)	Core entity attribute
platformFee	Decimal	@db.Decimal(10, 2)	Core entity attribute
currency	String	@default("INR")	Core entity attribute
status	PaymentStatus	@default(PENDING)	State machine lifecycle enum
provider	String	@default("razorpay")	Core entity attribute
providerOrderId	String?	@unique	Unique constraint
providerPaymentId	String?	@unique	Unique constraint
transactionId	String?	@unique	Unique constraint
paidAt	DateTime?	—	Timestamp lifecycle marker
refundedAt	DateTime?	—	Timestamp lifecycle marker
refundReason	String?	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
Booking	Booking	@relation(fields: [bookingId], references: [id])	Foreign key relationship
Host	Host	@relation(fields: [hostId], references: [id])	Foreign key relationship
User	User	@relation(fields: [userId], references: [id])	Foreign key relationship
Indexes & Constraints: @@index([hostId]) @@index([status]) @@index([userId])			

Model: PaymentWebhookEvent

10 Attributes • 2 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
provider	String	@default("razorpay")	Core entity attribute
eventId	String	@unique	Unique constraint
eventType	String	—	Core entity attribute
providerOrderId	String?	—	Core entity attribute
providerPaymentId	String?	—	Core entity attribute
status	String	@default("PROCESSING")	State machine lifecycle enum
failureReason	String?	—	Core entity attribute
receivedAt	DateTime	@default(now())	Timestamp lifecycle marker
processedAt	DateTime?	—	Timestamp lifecycle marker
Indexes & Constraints: @@index([providerOrderId]) @@index([status, receivedAt])			

Model: Payout

18 Attributes • 2 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
hostId	String	—	Core entity attribute
amount	Decimal	@db.Decimal(10, 2)	Core entity attribute
currency	String	@default("INR")	Core entity attribute
bankName	String?	—	Core entity attribute
accountNumber	String?	—	Core entity attribute
ifscCode	String?	—	Core entity attribute
status	PayoutStatus	@default(PENDING)	State machine lifecycle enum
transactionId	String?	—	Core entity attribute
requestedAt	DateTime	@default(now())	Timestamp lifecycle marker
processedAt	DateTime?	—	Timestamp lifecycle marker
processedById	String?	—	Core entity attribute
failureReason	String?	—	Core entity attribute
notes	String?	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
Host	Host	@relation(fields: [hostId], references: [id])	Foreign key relationship
User	User?	@relation(fields: [processedById], references: [id])	Foreign key relationship
Indexes & Constraints: @@index([hostId]) @@index([status])			

7.4 Domain 2: Statutory Compliance & Trust (7 Relational Models)

This domain encompasses 7 relational entities enforcing data integrity, foreign key cascades, and business constraints.

Model: AuditLog

16 Attributes • 5 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
userId	String?	—	Core entity attribute
action	String	—	Core entity attribute
entity	String	—	Core entity attribute
entityId	String?	—	Core entity attribute
oldData	Json?	—	Core entity attribute
newData	Json?	—	Core entity attribute
ipAddress	String?	—	Core entity attribute
userAgent	String?	—	Core entity attribute
severity	AuditSeverity	@default(INFO)	Core entity attribute
module	String?	—	Core entity attribute
requestId	String?	—	Core entity attribute
endpoint	String?	—	Core entity attribute
method	String?	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
User	User?	@relation(fields: [userId], references: [id])	Foreign key relationship
Indexes & Constraints: @@index([createdAt]) @@index([entity, entityId]) @@index([userId]) @@index([severity]) @@index([module])			

Model: Host

49 Attributes • 4 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
userId	String	@unique	Unique constraint
businessName	String?	—	Core entity attribute
businessType	BusinessType	@default(INDIVIDUAL)	Core entity attribute
hostType	HostType	@default(MULTI_SERVICE)	Core entity attribute
bio	String?	—	Core entity attribute
website	String?	—	Core entity attribute
logoUrl	String?	—	Core entity attribute
contactEmail	String?	—	Core entity attribute
supportPhone	String?	—	Core entity attribute
address	String?	—	Core entity attribute
city	String?	—	Core entity attribute
state	String?	—	Core entity attribute
country	String?	—	Core entity attribute
postalCode	String?	—	Core entity attribute
gstNumber	String?	—	Core entity attribute
registrationNo	String?	—	Core entity attribute
bankName	String?	—	Core entity attribute
bankAccountName	String?	—	Core entity attribute
bankAccountNumber	String?	—	Core entity attribute
bankIfsc	String?	—	Core entity attribute
isVerified	Boolean	@default(false)	Boolean state toggle
isApproved	Boolean	@default(false)	Boolean state toggle
isActive	Boolean	@default(true)	Boolean state toggle
kycStatus	KycStatus	@default(NOT_SUBMITTED)	State machine lifecycle enum
totalEarnings	Decimal	@default(0) @db.Decimal(12, 2)	Core entity attribute
averageRating	Float	@default(0)	Core entity attribute
joinedAt	DateTime	@default(now())	Timestamp lifecycle marker
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
reviewedAt	DateTime?	—	Timestamp lifecycle marker
reviewedById	String?	—	Core entity attribute
rejectionReason	String?	—	Core entity attribute
moderationNotes	String?	—	Core entity attribute
suspendedAt	DateTime?	—	Timestamp lifecycle marker
deactivatedAt	DateTime?	—	Timestamp lifecycle marker
Activity	Activity[]	—	Core entity attribute
ActivityBooking	ActivityBooking[]	—	Core entity attribute

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
Booking	Booking[]	—	Core entity attribute
User	User	@relation(fields: [userId], references: [id], onDelete: Cascade)	Foreign key relationship
ReviewedBy	User?	@relation("HostReviewer", fields: [reviewedById], references: [id])	Foreign key relationship
KycApplication	KycApplication[]	—	Core entity attribute
Payment	Payment[]	—	Core entity attribute
Payout	Payout[]	—	Core entity attribute
Rental	Rental[]	—	Core entity attribute
RentalBooking	RentalBooking[]	—	Core entity attribute
Review	Review[]	—	Core entity attribute
Tour	Tour[]	—	Core entity attribute
Incident	Incident[]	—	Core entity attribute
Indexes & Constraints: @@index([city]) @@index([isApproved, isActive]) @@index([kycStatus]) @@index([reviewedById])			

Model: KycApplication

70 Attributes • 3 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
hostId	String	—	Core entity attribute
status	KycStatus	@default(PENDING)	State machine lifecycle enum
firstName	String	—	Core entity attribute
lastName	String	—	Core entity attribute
dateOfBirth	DateTime?	—	Core entity attribute
nationality	String?	—	Core entity attribute
idType	String	—	Core entity attribute
idNumber	String	—	Core entity attribute
idFrontUrl	String?	—	Core entity attribute
idBackUrl	String?	—	Core entity attribute
addressProofUrl	String?	—	Core entity attribute
businessDocUrl	String?	—	Core entity attribute
submittedAt	DateTime	@default(now())	Timestamp lifecycle marker
reviewedById	String?	—	Core entity attribute
reviewedAt	DateTime?	—	Timestamp lifecycle marker
rejectionReason	String?	—	Core entity attribute
reviewNotes	String?	—	Core entity attribute
resubmissionAllowed	Boolean	@default(false)	Core entity attribute
resubmissionDeadline	DateTime?	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
Host	Host	@relation(fields: [hostId], references: [id], onDelete: Cascade)	Foreign key relationship
ReviewedBy	User?	@relation("KycReviewer", fields: [reviewedById], references: [id])	Foreign key relationship
id	String	@id @default(uuid())	Primary Key identifier
hostId	String	—	Core entity attribute
status	KycStatus	@default(PENDING)	State machine lifecycle enum
hostType	String	@default("INDIVIDUAL")	Core entity attribute
firstName	String	—	Core entity attribute
lastName	String	—	Core entity attribute
dateOfBirth	DateTime?	—	Core entity attribute
nationality	String?	—	Core entity attribute
idType	String	—	Core entity attribute
idNumber	String	—	Core entity attribute
idFrontUrl	String?	—	Core entity attribute
idBackUrl	String?	—	Core entity attribute

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
selfieUrl	String?	—	Core entity attribute
streetAddress	String?	—	Core entity attribute
city	String?	—	Core entity attribute
state	String?	—	Core entity attribute
postalCode	String?	—	Core entity attribute
country	String?	@default("India")	Core entity attribute
addressProofType	String?	—	Core entity attribute
addressProofUrl	String?	—	Core entity attribute
businessName	String?	—	Core entity attribute
gstIn	String?	—	Core entity attribute
businessPan	String?	—	Core entity attribute
businessDocUrl	String?	—	Core entity attribute
bankAccountName	String?	—	Core entity attribute
bankAccountNumber	String?	—	Core entity attribute
bankIfsc	String?	—	Core entity attribute
bankName	String?	—	Core entity attribute
cancelledChequeUrl	String?	—	Core entity attribute
emergencyContactName	String?	—	Core entity attribute
emergencyContactPhone	String?	—	Core entity attribute
emergencyContactRelation	String?	—	Core entity attribute
safetyCertUrl	String?	—	Core entity attribute
agreedToHostSafetyPolicy	Boolean	@default(false)	Core entity attribute
safetyPolicyAgreedAt	DateTime?	—	Timestamp lifecycle marker
submittedAt	DateTime	@default(now())	Timestamp lifecycle marker
reviewedById	String?	—	Core entity attribute
reviewedAt	DateTime?	—	Timestamp lifecycle marker
rejectionReason	String?	—	Core entity attribute
reviewNotes	String?	—	Core entity attribute
resubmissionAllowed	Boolean	@default(false)	Core entity attribute
resubmissionDeadline	DateTime?	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
Host	Host	@relation(fields: [hostId], references: [id], onDelete: Cascade)	Foreign key relationship
ReviewedBy	User?	@relation("KycReviewer", fields: [reviewedById], references: [id])	Foreign key relationship

Indexes & Constraints: @@index([hostId]) | @@index([status]) | @@index([reviewedById])

Model: TravelerReport

10 Attributes • 3 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
reporterId	String	—	Core entity attribute
reportedUserId	String	—	Core entity attribute
tourId	String	—	Core entity attribute
reason	String	—	Core entity attribute
details	String?	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
reporter	User	@relation("Reporter", fields: [reporterId], references: [id])	Foreign key relationship
reportedUser	User	@relation("ReportedUser", fields: [reportedUserId], references: [id])	Foreign key relationship
Tour	Tour	@relation(fields: [tourId], references: [id])	Foreign key relationship
Indexes & Constraints: @@index([tourId]) @@index([reporterId]) @@index([reportedUserId])			

Model: ModerationLog

12 Attributes • 4 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
entityType	ModerationEntityType	—	Core entity attribute
entityId	String	—	Core entity attribute
action	ModerationAction	—	Core entity attribute
oldStatus	String?	—	State machine lifecycle enum
newStatus	String?	—	State machine lifecycle enum
reason	String?	—	Core entity attribute
notes	String?	—	Core entity attribute
metadata	Json?	—	Core entity attribute
adminId	String	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
Admin	User	@relation("ModerationLogAdmin", fields: [adminId], references: [id])	Foreign key relationship
Indexes & Constraints: @@index([entityType, entityId]) @@index([adminId]) @@index([action]) @@index([createdAt])			

Model: Policy

13 Attributes • 3 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
type	PolicyType	—	Core entity attribute
title	String	—	Core entity attribute
version	String	—	Core entity attribute
summary	String?	@db.Text	Core entity attribute
content	String	@db.Text	Core entity attribute
isActive	Boolean	@default(true)	Boolean state toggle
effectiveDate	DateTime	@default(now())	Core entity attribute
publishedById	String?	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
PublishedBy	User?	@relation("PolicyPublisher", fields: [publishedById], references: [id])	Foreign key relationship
Consents	PolicyConsent[]	—	Core entity attribute

Indexes & Constraints: @@unique([type, version]) | @@index([type, isActive]) | @@index([publishedById])

Model: PolicyConsent

11 Attributes • 3 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
userId	String	—	Core entity attribute
policyId	String	—	Core entity attribute
policyType	PolicyType	—	Core entity attribute
policyVersion	String	—	Core entity attribute
ipAddress	String?	—	Core entity attribute
userAgent	String?	—	Core entity attribute
context	String	@default("SIGNUP")	Core entity attribute
consentedAt	DateTime	@default(now())	Timestamp lifecycle marker
User	User	@relation(fields: [userId], references: [id], onDelete: Cascade)	Foreign key relationship
Policy	Policy	@relation(fields: [policyId], references: [id], onDelete: Cascade)	Foreign key relationship

Indexes & Constraints: @@index([userId, policyType]) | @@index([policyId]) | @@index([consentedAt])

7.5 Domain 3: Multi-Day Guided Tours & Itineraries (16 Relational Models)

This domain encompasses 16 relational entities enforcing data integrity, foreign key cascades, and business constraints.

Model: Booking

39 Attributes • 6 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
bookingCode	String	@unique	Unique constraint
userId	String	—	Core entity attribute
hostId	String	—	Core entity attribute
tourId	String?	—	Core entity attribute
checkIn	DateTime?	—	Core entity attribute
checkOut	DateTime?	—	Core entity attribute
totalGuests	Int	@default(1)	Core entity attribute
adults	Int	@default(1)	Core entity attribute
children	Int	@default(0)	Core entity attribute
infants	Int	@default(0)	Core entity attribute
contactName	String	—	Core entity attribute
contactEmail	String	—	Core entity attribute
contactPhone	String	—	Core entity attribute
specialRequests	String?	—	Core entity attribute
riskAcknowledgedAt	DateTime?	—	Timestamp lifecycle marker
riskDisclosureSnapshot	Json?	—	Core entity attribute
couponId	String?	—	Core entity attribute
TourParticipant	TourParticipant[]	—	Core entity attribute
subtotal	Decimal	@db.Decimal(10, 2)	Core entity attribute
taxes	Decimal	@default(0) @db.Decimal(10, 2)	Core entity attribute
discount	Decimal	@default(0) @db.Decimal(10, 2)	Core entity attribute
totalAmount	Decimal	@db.Decimal(10, 2)	Monetary currency value (INR)
currency	String	@default("INR")	Core entity attribute
status	BookingStatus	@default(PENDING)	State machine lifecycle enum
expiresAt	DateTime?	—	Timestamp lifecycle marker
cancellationReason	String?	—	Core entity attribute
cancelledAt	DateTime?	—	Timestamp lifecycle marker
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
Coupon	Coupon?	@relation(fields: [couponId], references: [id])	Foreign key relationship
Host	Host	@relation(fields: [hostId], references: [id])	Foreign key relationship
Tour	Tour?	@relation(fields: [tourId], references: [id], onDelete: Restrict)	Foreign key relationship
User	User	@relation(fields: [userId], references: [id])	Foreign key relationship
BookingTimeline	BookingTimeline[]	—	Core entity attribute
CouponRedemption	CouponRedemption[]	—	Core entity attribute
Payment	Payment?	—	Core entity attribute

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
Review	Review?	—	Core entity attribute
Incident	Incident[]	—	Core entity attribute

Indexes & Constraints: @@index([bookingCode]) | @@index([hostId]) | @@index([status]) | @@index([checkIn, checkOut]) | @@index([tourId]) | @@index([userId])

Model: BookingTimeline

8 Attributes • 1 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
bookingId	String	—	Core entity attribute
type	BookingEventType	—	Core entity attribute
title	String	—	Core entity attribute
message	String?	—	Core entity attribute
metadata	Json?	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
Booking	Booking	@relation(fields: [bookingId], references: [id], onDelete: Cascade)	Foreign key relationship

Indexes & Constraints: @@index([bookingId])

Model: TourMessageReport

18 Attributes • 4 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
reporterId	String	—	Core entity attribute
messageId	String	—	Core entity attribute
tourId	String	—	Core entity attribute
reportedUserId	String	—	Core entity attribute
reason	CommunityReportReason	—	Core entity attribute
details	String?	—	Core entity attribute
status	CommunityReportStatus	@default(OPEN)	State machine lifecycle enum
resolutionNotes	String?	—	Core entity attribute
reviewedById	String?	—	Core entity attribute
reviewedAt	DateTime?	—	Timestamp lifecycle marker
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
Reporter	User	@relation("TourMessageReportReporter", fields: [reporterId], references: [id], onDelete: Restrict)	Foreign key relationship
ReportedUser	User	@relation("TourMessageReportSubject", fields: [reportedUserId], references: [id], onDelete: Restrict)	Foreign key relationship
Reviewer	User?	@relation("TourMessageReportReviewer", fields: [reviewedById], references: [id], onDelete: SetNull)	Foreign key relationship
Message	TourMessage	@relation(fields: [messageId], references: [id], onDelete: Restrict)	Foreign key relationship
Tour	Tour	@relation(fields: [tourId], references: [id], onDelete: Restrict)	Foreign key relationship

Indexes & Constraints: @@unique([reporterId, messageId]) | @@index([status, createdAt]) | @@index([tourId]) | @@index([reportedUserId])

Model: Tour

73 Attributes • 6 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
hostId	String	—	Core entity attribute
slug	String	@unique	Unique constraint
title	String	—	Core entity attribute
description	String	—	Core entity attribute
destination	String	—	Core entity attribute
city	String?	—	Core entity attribute
state	String?	—	Core entity attribute
joinApprovalRequired	Boolean	@default(false)	Core entity attribute
country	String?	—	Core entity attribute
latitude	Float?	—	Core entity attribute
longitude	Float?	—	Core entity attribute
duration	Int	—	Core entity attribute
tourStatus	TourStatus	@default(UPCOMING)	State machine lifecycle enum
maxGroupSize	Int	@default(20)	Core entity attribute
totalSlots	Int	@default(20)	Core entity attribute
availableSlots	Int	@default(20)	Core entity attribute
pricePerPerson	Decimal	@db.Decimal(10, 2)	Monetary currency value (INR)
originalPrice	Decimal?	@db.Decimal(10, 2)	Core entity attribute
difficulty	TourDifficulty	@default(MODERATE)	Core entity attribute
riskLevel	TourRiskLevel	@default(LOW)	Core entity attribute
riskDisclosure	String?	—	Core entity attribute
meetingPoint	String?	—	Core entity attribute
eligibilityRequirements	String[]	@default([])	Core entity attribute
requiredEquipment	String[]	@default([])	Core entity attribute
emergencyPlan	String?	—	Core entity attribute
minimumAge	Int	@default(18)	Core entity attribute
requiresCaretaker	Boolean	@default(false)	Core entity attribute
category	String?	—	Core entity attribute
tags	String[]	—	Core entity attribute
images	String[]	—	Core entity attribute
highlights	String[]	—	Core entity attribute
chatRoom	TourChatRoom?	—	Core entity attribute
included	String[]	—	Core entity attribute
womenOnly	Boolean	@default(false)	Core entity attribute
safeForSoloWomen	Boolean	@default(false)	Core entity attribute
verifiedTravelersOnly	Boolean	@default(false)	Core entity attribute
excluded	String[]	—	Core entity attribute

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
languages	String[]	—	Core entity attribute
cancellationPolicy	String?	—	Core entity attribute
status	ListingStatus	@default(PENDING_REVIEW)	State machine lifecycle enum
isActive	Boolean	@default(true)	Boolean state toggle
isApproved	Boolean	@default(false)	Boolean state toggle
averageRating	Float	@default(0)	Core entity attribute
totalReviews	Int	@default(0)	Core entity attribute
totalBookings	Int	@default(0)	Core entity attribute
submittedForReviewAt	DateTime?	—	Timestamp lifecycle marker
approvedAt	DateTime?	—	Timestamp lifecycle marker
rejectedAt	DateTime?	—	Timestamp lifecycle marker
archivedAt	DateTime?	—	Timestamp lifecycle marker
deletedAt	DateTime?	—	Timestamp lifecycle marker
reviewedAt	DateTime?	—	Timestamp lifecycle marker
reviewedBy	String?	—	Core entity attribute
rejectionReason	String?	—	Core entity attribute
moderationNotes	String?	—	Core entity attribute
lastReviewedStatus	ListingStatus?	—	State machine lifecycle enum
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
Booking	Booking[]	—	Core entity attribute
Post	Post[]	—	Core entity attribute
startDate	DateTime	—	Core entity attribute
endDate	DateTime	—	Core entity attribute
registrationDeadline	DateTime?	—	Core entity attribute
Review	Review[]	—	Core entity attribute
Host	Host	@relation(fields: [hostId], references: [id], onDelete: Cascade)	Foreign key relationship
ReviewedBy	User?	@relation("TourReviewer", fields: [reviewedBy], references: [id])	Foreign key relationship
TourItineraryDay	TourItineraryDay[]	—	Core entity attribute
TourParticipant	TourParticipant[]	—	Core entity attribute
TravelerReport	TravelerReport[]	—	Core entity attribute
TourMessageReport	TourMessageReport[]	—	Core entity attribute
TourWaitlist	TourWaitlist[]	—	Core entity attribute
WishlistItem	WishlistItem[]	—	Core entity attribute
Incident	Incident[]	—	Core entity attribute

Indexes & Constraints: @@index([city]) | @@index([hostId]) | @@index([status, isActive, isApproved]) | @@index([riskLevel, status]) | @@index([reviewedBy]) | @@index([deletedAt])

Model: TourItineraryDay

10 Attributes • 1 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
tourId	String	—	Core entity attribute
day	Int	—	Core entity attribute
title	String	—	Core entity attribute
description	String	—	Core entity attribute
activities	String[]	—	Core entity attribute
meals	String[]	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
Tour	Tour	@relation(fields: [tourId], references: [id], onDelete: Cascade)	Foreign key relationship

Indexes & Constraints: @@unique([tourId, day])

Model: TourJoinRequest

6 Attributes • 0 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
userId	String	—	Core entity attribute
tourId	String	—	Core entity attribute
introduction	String?	—	Core entity attribute
status	JoinRequestStatus	@default(PENDING)	State machine lifecycle enum
createdAt	DateTime	@default(now())	Timestamp lifecycle marker

Model: TourMessageSeen

6 Attributes • 1 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
messageId	String	—	Core entity attribute
userId	String	—	Core entity attribute
seenAt	DateTime	@default(now())	Timestamp lifecycle marker
Message	TourMessage	@relation(fields: [messageId], references: [id], onDelete: Cascade)	Foreign key relationship
User	User	@relation(fields: [userId], references: [id], onDelete: Cascade)	Foreign key relationship

Indexes & Constraints: @@unique([messageId, userId])

Model: TourChatRoom

10 Attributes • 0 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
tourId	String	@unique	Unique constraint
name	String?	—	Core entity attribute
imageUrl	String?	—	Core entity attribute
isActive	Boolean	@default(true)	Boolean state toggle
lastMessage	String?	—	Core entity attribute
lastMessageAt	DateTime?	—	Timestamp lifecycle marker
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
Tour	Tour	@relation(fields: [tourId], references: [id])	Foreign key relationship
messages	TourMessage[]	—	Core entity attribute

Model: TourMessage

17 Attributes • 2 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
roomId	String	—	Core entity attribute
senderId	String	—	Core entity attribute
participantId	String?	—	Core entity attribute
message	String?	—	Core entity attribute
imageUrl	String?	—	Core entity attribute
messageType	MessageType	@default(TEXT)	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
deletedAt	DateTime?	—	Timestamp lifecycle marker
isEdited	Boolean	@default(false)	Boolean state toggle
editedAt	DateTime?	—	Timestamp lifecycle marker
replyToId	String?	—	Core entity attribute
TourChatRoom	TourChatRoom	@relation(fields: [roomId], references: [id])	Foreign key relationship
User	User	@relation(fields: [senderId], references: [id])	Foreign key relationship
Participant	TourParticipant?	@relation(fields: [participantId], references: [id])	Foreign key relationship
SeenBy	TourMessageSeen[]	—	Core entity attribute
Reports	TourMessageReport[]	—	Core entity attribute

Indexes & Constraints: @@index([roomId, createdAt]) | @@index([senderId])

Model: TourWaitlist

7 Attributes • 0 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
userId	String	—	Core entity attribute
tourId	String	—	Core entity attribute
position	Int	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
User	User	@relation(fields: [userId], references: [id])	Foreign key relationship
Tour	Tour	@relation(fields: [tourId], references: [id])	Foreign key relationship

Model: TourBooking

28 Attributes • 4 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
bookingCode	String	@unique	Unique constraint
userId	String	—	Core entity attribute
hostId	String	—	Core entity attribute
tourId	String	—	Core entity attribute
batchId	String?	—	Core entity attribute
legacyBookingId	String?	@unique	Unique constraint
idempotencyKey	String?	@unique	Unique constraint
status	String	@default("PENDING")	State machine lifecycle enum
paymentStatus	String	@default("PENDING")	State machine lifecycle enum
travelersCount	Int	@default(1)	Core entity attribute
confirmedCount	Int	@default(0)	Core entity attribute
waitlistedCount	Int	@default(0)	Core entity attribute
unitPrice	Decimal	@db.Decimal(10, 2)	Core entity attribute
subtotal	Decimal	@db.Decimal(10, 2)	Core entity attribute
taxes	Decimal	@default(0) @db.Decimal(10, 2)	Core entity attribute
discount	Decimal	@default(0) @db.Decimal(10, 2)	Core entity attribute
totalAmount	Decimal	@db.Decimal(10, 2)	Monetary currency value (INR)
currency	String	@default("INR")	Core entity attribute
expiresAt	DateTime?	—	Timestamp lifecycle marker
confirmedAt	DateTime?	—	Timestamp lifecycle marker
cancelledAt	DateTime?	—	Timestamp lifecycle marker
deletedAt	DateTime?	—	Timestamp lifecycle marker
metadata	Json?	—	Core entity attribute
riskAcknowledgedAt	DateTime?	—	Timestamp lifecycle marker
riskDisclosureSnapshot	Json?	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker

Indexes & Constraints: @@index([userId, createdAt]) | @@index([tourId, status]) | @@index([batchId, status]) | @@index([hostId, status])

Model: TourTraveler

25 Attributes • 5 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
tourBookingId	String	—	Core entity attribute
userId	String?	—	Core entity attribute
fullName	String	—	Core entity attribute
normalizedName	String?	—	Core entity attribute
age	Int?	—	Core entity attribute
dob	DateTime?	—	Core entity attribute
aadhaarHash	String?	—	Core entity attribute
aadhaarLast4	String?	—	Core entity attribute
gender	String?	—	Core entity attribute
email	String?	—	Core entity attribute
phone	String?	—	Core entity attribute
emergencyContactName	String?	—	Core entity attribute
emergencyContactPhone	String?	—	Core entity attribute
country	String?	—	Core entity attribute
foodPreference	String?	—	Core entity attribute
medicalNotes	String?	—	Core entity attribute
bloodGroup	String?	—	Core entity attribute
idType	String?	—	Core entity attribute
idUploadUrl	String?	—	Core entity attribute
relation	String?	—	Core entity attribute
seatPreference	String?	—	Core entity attribute
status	String	@default("PENDING")	State machine lifecycle enum
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker

Indexes & Constraints: @@index([tourBookingId, status]) | @@index([email, phone]) | @@index([aadhaarHash]) | @@index([normalizedName, age]) | @@index([normalizedName, dob])

Model: **WaitlistQueue**

13 Attributes • 1 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
tourId	String	—	Core entity attribute
batchId	String?	—	Core entity attribute
tourBookingId	String	@unique	Unique constraint
userId	String	—	Core entity attribute
groupSize	Int	@default(1)	Core entity attribute
position	Int	—	Core entity attribute
status	String	@default("WAITLISTED")	State machine lifecycle enum
promotedAt	DateTime?	—	Timestamp lifecycle marker
expiresAt	DateTime?	—	Timestamp lifecycle marker
declinedAt	DateTime?	—	Timestamp lifecycle marker
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
Indexes & Constraints: @@index([tourId, status, position])			

Model: **TourCancellation**

11 Attributes • 1 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
tourBookingId	String	—	Core entity attribute
travelerId	String?	—	Core entity attribute
cancelledById	String	—	Core entity attribute
scope	String	@default("BOOKING")	Core entity attribute
reason	String?	—	Core entity attribute
refundPercent	Int	@default(0)	Core entity attribute
refundAmount	Decimal	@default(0) @db.Decimal(10, 2)	Monetary currency value (INR)
status	String	@default("REQUESTED")	State machine lifecycle enum
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
processedAt	DateTime?	—	Timestamp lifecycle marker
Indexes & Constraints: @@index([tourBookingId, status])			

Model: Refund

11 Attributes • 1 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
tourBookingId	String	—	Core entity attribute
paymentId	String?	—	Core entity attribute
amount	Decimal	@db.Decimal(10, 2)	Core entity attribute
currency	String	@default("INR")	Core entity attribute
status	String	@default("PENDING")	State machine lifecycle enum
reason	String?	—	Core entity attribute
providerRefundId	String?	—	Core entity attribute
failureReason	String?	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
processedAt	DateTime?	—	Timestamp lifecycle marker
Indexes & Constraints: @@index([tourBookingId, status])			

Model: TourParticipant

17 Attributes • 3 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
tourId	String	—	Core entity attribute
userId	String	—	Core entity attribute
bookingId	String?	—	Core entity attribute
status	TourParticipantStatus	@default(PENDING)	State machine lifecycle enum
joinedAt	DateTime	@default(now())	Timestamp lifecycle marker
introMessage	String?	—	Core entity attribute
role	TourRole	@default(MEMBER)	Core entity attribute
isVerified	Boolean	@default(false)	Boolean state toggle
isHostApproved	Boolean	@default(false)	Boolean state toggle
Tour	Tour	@relation(fields: [tourId], references: [id], onDelete: Cascade)	Foreign key relationship
User	User	@relation(fields: [userId], references: [id], onDelete: Cascade)	Foreign key relationship
Booking	Booking?	@relation(fields: [bookingId], references: [id])	Foreign key relationship
TourMessage	TourMessage[]	—	Core entity attribute
checkedInAt	DateTime?	—	Timestamp lifecycle marker
completedAt	DateTime?	—	Timestamp lifecycle marker
cancelledAt	DateTime?	—	Timestamp lifecycle marker
Indexes & Constraints: @@unique([tourId, userId]) @@index([tourId]) @@index([userId])			

7.6 Domain 7: Community, Reviews, Social & Safety (9 Relational Models)

This domain encompasses 9 relational entities enforcing data integrity, foreign key cascades, and business constraints.

Model: Notification

11 Attributes • 2 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
userId	String	—	Core entity attribute
type	NotificationType	—	Core entity attribute
title	String	—	Core entity attribute
message	String	—	Core entity attribute
data	Json?	—	Core entity attribute
isRead	Boolean	@default(false)	Boolean state toggle
readAt	DateTime?	—	Timestamp lifecycle marker
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
User	User	@relation(fields: [userId], references: [id], onDelete: Cascade)	Foreign key relationship
NotificationDelivery	NotificationDelivery[]	—	Core entity attribute
Indexes & Constraints: @@index([userId, createdAt]) @@index([userId, isRead])			

Model: NotificationDelivery

14 Attributes • 3 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
notificationId	String	—	Core entity attribute
userId	String	—	Core entity attribute
channel	NotificationDeliveryChannel	@default(EMAIL)	Core entity attribute
status	NotificationDeliveryStatus	@default(PENDING)	State machine lifecycle enum
attempts	Int	@default(0)	Core entity attribute
availableAt	DateTime	@default(now())	Timestamp lifecycle marker
lockedAt	DateTime?	—	Timestamp lifecycle marker
deliveredAt	DateTime?	—	Timestamp lifecycle marker
lastError	String?	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
Notification	Notification	@relation(fields: [notificationId], references: [id], onDelete: Cascade)	Foreign key relationship
User	User	@relation(fields: [userId], references: [id], onDelete: Cascade)	Foreign key relationship
Indexes & Constraints: @@unique([notificationId, channel]) @@index([status, availableAt]) @@index([userId, status])			

Model: **Post**

12 Attributes • 1 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
userId	String	—	Core entity attribute
tourId	String?	—	Core entity attribute
imageUrl	String	—	Core entity attribute
caption	String?	—	Core entity attribute
viewCount	Int	@default(0)	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
Tour	Tour?	@relation(fields: [tourId], references: [id])	Foreign key relationship
User	User	@relation(fields: [userId], references: [id], onDelete: Cascade)	Foreign key relationship
PostComment	PostComment[]	—	Core entity attribute
PostLike	PostLike[]	—	Core entity attribute
Indexes & Constraints: @@index([userId])			

Model: **PostComment**

8 Attributes • 2 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
postId	String	—	Core entity attribute
userId	String	—	Core entity attribute
content	String	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
Post	Post	@relation(fields: [postId], references: [id], onDelete: Cascade)	Foreign key relationship
User	User	@relation(fields: [userId], references: [id], onDelete: Cascade)	Foreign key relationship
Indexes & Constraints: @@index([postId]) @@index([userId])			

Model: PostLike

6 Attributes • 1 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
postId	String	—	Core entity attribute
userId	String	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
Post	Post	@relation(fields: [postId], references: [id], onDelete: Cascade)	Foreign key relationship
User	User	@relation(fields: [userId], references: [id], onDelete: Cascade)	Foreign key relationship
Indexes & Constraints: @@unique([postId, userId])			

Model: Review

26 Attributes • 4 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
userId	String	—	Core entity attribute
hostId	String?	—	Core entity attribute
tourId	String?	—	Core entity attribute
activityId	String?	—	Core entity attribute
rentalId	String?	—	Core entity attribute
bookingId	String?	@unique	Unique constraint
activityBookingId	String?	@unique	Unique constraint
rentalBookingId	String?	@unique	Unique constraint
target	ReviewTarget	—	Core entity attribute
rating	Int	—	Core entity attribute
title	String?	—	Core entity attribute
comment	String	—	Core entity attribute
response	String?	—	Core entity attribute
responseAt	DateTime?	—	Timestamp lifecycle marker
isPublished	Boolean	@default(true)	Boolean state toggle
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
ActivityBooking	ActivityBooking?	@relation(fields: [activityBookingId], references: [id])	Foreign key relationship
Activity	Activity?	@relation(fields: [activityId], references: [id])	Foreign key relationship
Booking	Booking?	@relation(fields: [bookingId], references: [id])	Foreign key relationship
Host	Host?	@relation(fields: [hostId], references: [id])	Foreign key relationship
RentalBooking	RentalBooking?	@relation(fields: [rentalBookingId], references: [id])	Foreign key relationship
Rental	Rental?	@relation(fields: [rentalId], references: [id])	Foreign key relationship
Tour	Tour?	@relation(fields: [tourId], references: [id])	Foreign key relationship
User	User	@relation(fields: [userId], references: [id], onDelete: Cascade)	Foreign key relationship

Indexes & Constraints: @@index([activityId]) | @@index([rentalId]) | @@index([tourId]) | @@index([userId])

Model: WishlistItem

11 Attributes • 4 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
userId	String	—	Core entity attribute
target	WishlistTarget	—	Core entity attribute
tourId	String?	—	Core entity attribute
activityId	String?	—	Core entity attribute
rentalId	String?	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
Activity	Activity?	@relation(fields: [activityId], references: [id])	Foreign key relationship
Rental	Rental?	@relation(fields: [rentalId], references: [id])	Foreign key relationship
Tour	Tour?	@relation(fields: [tourId], references: [id])	Foreign key relationship
User	User	@relation(fields: [userId], references: [id], onDelete: Cascade)	Foreign key relationship

Indexes & Constraints: @@unique([userId, activityId]) | @@unique([userId, rentalId]) | @@unique([userId, tourId]) | @@index([userId])

Model: Incident

25 Attributes • 6 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
referenceCode	String	@unique	Unique constraint
reporterId	String	—	Core entity attribute
ownerId	String?	—	Core entity attribute
bookingId	String	—	Core entity attribute
tourId	String	—	Core entity attribute
hostId	String	—	Core entity attribute
category	IncidentCategory	—	Core entity attribute
severity	IncidentSeverity	@default(MEDIUM)	Core entity attribute
status	IncidentStatus	@default(OPEN)	State machine lifecycle enum
title	String	—	Core entity attribute
description	String	—	Core entity attribute
immediateDanger	Boolean	@default(false)	Core entity attribute
resolutionSummary	String?	—	Core entity attribute
triagedAt	DateTime?	—	Timestamp lifecycle marker
resolvedAt	DateTime?	—	Timestamp lifecycle marker
closedAt	DateTime?	—	Timestamp lifecycle marker
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
Reporter	User	@relation("IncidentReporter", fields: [reporterId], references: [id], onDelete: Restrict)	Foreign key relationship
Owner	User?	@relation("IncidentOwner", fields: [ownerId], references: [id], onDelete: SetNull)	Foreign key relationship
Booking	Booking	@relation(fields: [bookingId], references: [id], onDelete: Restrict)	Foreign key relationship
Tour	Tour	@relation(fields: [tourId], references: [id], onDelete: Restrict)	Foreign key relationship
Host	Host	@relation(fields: [hostId], references: [id], onDelete: Restrict)	Foreign key relationship
IncidentEvent	IncidentEvent[]	—	Core entity attribute

Indexes & Constraints: @@index([reporterId, createdAt]) | @@index([ownerId, status]) | @@index([status, severity, createdAt]) | @@index([tourId]) | @@index([hostId]) | @@index([bookingId])

Model: IncidentEvent

11 Attributes • 2 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
incidentId	String	—	Core entity attribute
actorId	String	—	Core entity attribute
action	String	—	Core entity attribute
note	String?	—	Core entity attribute
oldStatus	IncidentStatus?	—	State machine lifecycle enum
newStatus	IncidentStatus?	—	State machine lifecycle enum
metadata	Json?	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
Incident	Incident	@relation(fields: [incidentId], references: [id], onDelete: Cascade)	Foreign key relationship
Actor	User	@relation("IncidentActor", fields: [actorId], references: [id], onDelete: Restrict)	Foreign key relationship
Indexes & Constraints: @@index([incidentId, createdAt]) @@index([actorId])			

7.7 Domain 5: Vehicle Fleet Rentals & Custody Inspections (7 Relational Models)

This domain encompasses 7 relational entities enforcing data integrity, foreign key cascades, and business constraints.

Model: Rental 6 Attributes • 0 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
hostId	String	—	Core entity attribute
slug	String	@unique	Unique constraint
title	String	—	Core entity attribute
vehicleType	VehicleType	—	Core entity attribute
brand	String	—	Core entity attribute

Model: RentalBooking

35 Attributes • 5 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
bookingCode	String	@unique	Unique constraint
idempotencyKey	String?	@unique	Unique constraint
userId	String	—	Core entity attribute
hostId	String	—	Core entity attribute
rentalId	String	—	Core entity attribute
pickupDate	DateTime	—	Core entity attribute
returnDate	DateTime	—	Core entity attribute
pickupTime	String?	—	Core entity attribute
days	Int	—	Core entity attribute
withHelmet	Boolean	@default(false)	Core entity attribute
withDelivery	Boolean	@default(false)	Core entity attribute
deliveryFee	Decimal	@default(0) @db.Decimal(10, 2)	Core entity attribute
subtotal	Decimal	@db.Decimal(10, 2)	Core entity attribute
taxes	Decimal	@default(0) @db.Decimal(10, 2)	Core entity attribute
totalAmount	Decimal	@db.Decimal(10, 2)	Monetary currency value (INR)
currency	String	@default("INR")	Core entity attribute
status	BookingStatus	@default(PENDING)	State machine lifecycle enum
expiresAt	DateTime?	—	Timestamp lifecycle marker
contactName	String	—	Core entity attribute
contactEmail	String	—	Core entity attribute
contactPhone	String	—	Core entity attribute
notes	String?	—	Core entity attribute
cancellationReason	String?	—	Core entity attribute
cancelledAt	DateTime?	—	Timestamp lifecycle marker
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
Host	Host	@relation(fields: [hostId], references: [id])	Foreign key relationship
Rental	Rental	@relation(fields: [rentalId], references: [id])	Foreign key relationship
User	User	@relation(fields: [userId], references: [id])	Foreign key relationship
RentalBookingTimeline	RentalBookingTimeline[]	—	Core entity attribute
RentalInspection	RentalInspection[]	—	Core entity attribute
RentalPayment	RentalPayment?	—	Core entity attribute
RentalRefund	RentalRefund?	—	Core entity attribute
Review	Review?	—	Core entity attribute

Indexes & Constraints: @@index([hostId]) | @@index([rentalId]) | @@index([status]) | @@index([expiresAt]) | @@index([userId])

Model: RentalInspection

24 Attributes • 3 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
bookingId	String	—	Core entity attribute
stage	RentalInspectionStage	—	Core entity attribute
status	RentalInspectionStatus	@default(DRAFT)	State machine lifecycle enum
odometerKm	Int?	—	Core entity attribute
fuelOrChargePercent	Int?	—	Core entity attribute
conditionNotes	String?	—	Core entity attribute
checklist	Json?	—	Core entity attribute
damages	Json?	—	Core entity attribute
evidenceAssetIds	String[]	—	Core entity attribute
recordedByUserId	String	—	Core entity attribute
submittedAt	DateTime?	—	Timestamp lifecycle marker
acknowledgedByUserId	String?	—	Core entity attribute
acknowledgedAt	DateTime?	—	Timestamp lifecycle marker
disputedByUserId	String?	—	Core entity attribute
disputedAt	DateTime?	—	Timestamp lifecycle marker
disputeReason	String?	—	Core entity attribute
resolution	RentalDisputeResolution?	—	Core entity attribute
resolutionNotes	String?	—	Core entity attribute
resolvedByUserId	String?	—	Core entity attribute
resolvedAt	DateTime?	—	Timestamp lifecycle marker
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
RentalBooking	RentalBooking	@relation(fields: [bookingId], references: [id], onDelete: Cascade)	Foreign key relationship

Indexes & Constraints: @@unique([bookingId, stage]) | @@index([bookingId, status]) | @@index([submittedAt])

Model: RentalBookingTimeline

8 Attributes • 1 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
bookingId	String	—	Core entity attribute
type	BookingEventType	—	Core entity attribute
title	String	—	Core entity attribute
message	String?	—	Core entity attribute
metadata	Json?	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
RentalBooking	RentalBooking	@relation(fields: [bookingId], references: [id], onDelete: Cascade)	Foreign key relationship

Indexes & Constraints: @@index([bookingId])

Model: RentalDetails

13 Attributes • 0 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
rentalId	String	@unique	Unique constraint
transmission	TransmissionType?	—	Core entity attribute
fuelType	FuelType?	—	Core entity attribute
seats	Int?	—	Core entity attribute
engine	String?	—	Core entity attribute
rangeKm	Int	@default(0)	Core entity attribute
deposit	Decimal	@default(0) @db.Decimal(10, 2)	Core entity attribute
features	String[]	—	Core entity attribute
documentsRequired	String[]	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
Rental	Rental	@relation(fields: [rentalId], references: [id], onDelete: Cascade)	Foreign key relationship

Model: RentalPayment

15 Attributes • 1 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
rentalBookingId	String	@unique	Unique constraint
userId	String	—	Core entity attribute
amount	Decimal	@db.Decimal(10, 2)	Core entity attribute
currency	String	@default("INR")	Core entity attribute
status	PaymentStatus	@default(PENDING)	State machine lifecycle enum
provider	String	@default("razorpay")	Core entity attribute
providerOrderId	String?	@unique	Unique constraint
providerPaymentId	String?	@unique	Unique constraint
transactionId	String?	@unique	Unique constraint
paidAt	DateTime?	—	Timestamp lifecycle marker
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
RentalBooking	RentalBooking	@relation(fields: [rentalBookingId], references: [id], onDelete: Cascade)	Foreign key relationship
RentalRefund	RentalRefund?	—	Core entity attribute

Indexes & Constraints: @@index([status])

Model: RentalRefund

15 Attributes • 1 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
id	String	@id @default(uuid())	Primary Key identifier
rentalBookingId	String	@unique	Unique constraint
rentalPaymentId	String?	@unique	Unique constraint
requestedAmount	Decimal	@db.Decimal(10, 2)	Monetary currency value (INR)
approvedAmount	Decimal?	@db.Decimal(10, 2)	Monetary currency value (INR)
currency	String	@default("INR")	Core entity attribute
status	String	@default("REVIEW_PENDING")	State machine lifecycle enum
reason	String?	—	Core entity attribute
policySnapshot	String	—	Core entity attribute
providerRefundId	String?	—	Core entity attribute
failureReason	String?	—	Core entity attribute
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
processedAt	DateTime?	—	Timestamp lifecycle marker
RentalBooking	RentalBooking	@relation(fields: [rentalBookingId], references: [id], onDelete: Cascade)	Foreign key relationship
RentalPayment	RentalPayment?	@relation(fields: [rentalPaymentId], references: [id])	Foreign key relationship

Indexes & Constraints: @@index([status, createdAt])

7.8 Domain 8: Auxiliary & Platform Utilities (1 Relational Models)

This domain encompasses 1 relational entities enforcing data integrity, foreign key cascades, and business constraints.

Model: String?

36 Attributes • 5 Indexes

FIELD NAME	DATA TYPE	CONSTRAINTS / RELATIONS / ATTRIBUTES	DOMAIN PURPOSE
year	Int?	—	Core entity attribute
city	String	—	Core entity attribute
state	String?	—	Core entity attribute
country	String	@default("India")	Core entity attribute
pickupArea	String?	—	Core entity attribute
totalUnits	Int	@default(1)	Core entity attribute
availableUnits	Int	@default(1)	Core entity attribute
imageUrl	String?	—	Core entity attribute
images	String[]	—	Core entity attribute
pricePerDay	Decimal	@db.Decimal(10, 2)	Monetary currency value (INR)
originalPrice	Decimal?	@db.Decimal(10, 2)	Core entity attribute
cancellationPolicy	String?	—	Core entity attribute
status	ListingStatus	@default(PENDING_REVIEW)	State machine lifecycle enum
isActive	Boolean	@default(true)	Boolean state toggle
isApproved	Boolean	@default(false)	Boolean state toggle
averageRating	Float	@default(0)	Core entity attribute
totalReviews	Int	@default(0)	Core entity attribute
totalBookings	Int	@default(0)	Core entity attribute
submittedForReviewAt	DateTime?	—	Timestamp lifecycle marker
approvedAt	DateTime?	—	Timestamp lifecycle marker
rejectedAt	DateTime?	—	Timestamp lifecycle marker
archivedAt	DateTime?	—	Timestamp lifecycle marker
deletedAt	DateTime?	—	Timestamp lifecycle marker
reviewedAt	DateTime?	—	Timestamp lifecycle marker
reviewedById	String?	—	Core entity attribute
rejectionReason	String?	—	Core entity attribute
moderationNotes	String?	—	Core entity attribute
lastReviewedStatus	ListingStatus?	—	State machine lifecycle enum
createdAt	DateTime	@default(now())	Timestamp lifecycle marker
updatedAt	DateTime	@updatedAt	Timestamp lifecycle marker
Host	Host	@relation(fields: [hostId], references: [id], onDelete: Cascade)	Foreign key relationship
ReviewedBy	User?	@relation("RentalReviewer", fields: [reviewedById], references: [id])	Foreign key relationship
RentalBooking	RentalBooking[]	—	Core entity attribute
RentalDetails	RentalDetails?	—	Core entity attribute
Review	Review[]	—	Core entity attribute
WishlistItem	WishlistItem[]	—	Core entity attribute

Indexes & Constraints: @@index([city, vehicleType]) | @@index([hostId]) | @@index([status, isActive, isApproved]) | @@index([reviewedById]) | @@index([deletedAt])

7.9 Complete Catalog of Strict Platform Enums (All 46 Enums)

PostgreSQL strict enums govern lifecycle states, user roles, policy categories, inspection stages, and severity tiers, preventing invalid state mutations across the application layer.

Enum: `TourStatus`

4 States

`UPCOMING` `ONGOING` `COMPLETED` `CANCELLED`**Enum:** `CommunityReportReason`

7 States

`HARASSMENT` `HATE_OR_ABUSE` `SEXUAL_CONTENT`
`THREAT_OR_SAFETY` `SCAM_OR_SPAM` `PRIVACY_VIOLATION`
`OTHER`**Enum:** `CommunityReportStatus`

4 States

`OPEN` `IN_REVIEW` `ACTIONED` `DISMISSED`**Enum:** `TravelStyle`

8 States

`BACKPACKER` `LUXURY` `ADVENTURE` `SOLO` `FAMILY`
`SPIRITUAL` `CULTURAL` `DIGITAL_NOMAD`**Enum:** `Gender`

4 States

`MALE` `FEMALE` `NON_BINARY` `PREFER_NOT_TO_SAY`**Enum:** `MessageType`

3 States

`TEXT` `IMAGE` `SYSTEM`**Enum:** `AccountStatus`

6 States

`PENDING` `ACTIVE` `SUSPENDED` `REJECTED` `BANNED`
`DELETED`**Enum:** `OtpPurpose`

5 States

`REGISTER` `LOGIN` `RESET_PASSWORD` `CHANGE_EMAIL`
`CHANGE_PHONE`**Enum:** `AuditSeverity`

5 States

`DEBUG` `INFO` `WARN` `ERROR` `CRITICAL`**Enum:** `IncidentCategory`

9 States

`SAFETY` `HARASSMENT` `MEDICAL` `TRANSPORT` `HOST_CONDUCT`
`TRAVELER_CONDUCT` `LOST_PERSON` `PROPERTY` `OTHER`**Enum:** `IncidentSeverity`

4 States

`LOW` `MEDIUM` `HIGH` `CRITICAL`**Enum:** `IncidentStatus`

5 States

`OPEN` `TRIAGED` `IN_PROGRESS` `RESOLVED` `CLOSED`**Enum:** `ModerationAction`

11 States

`APPROVED` `REJECTED` `SUSPENDED` `BANNED` `REINSTATED`
`DELETED` `ARCHIVED` `FLAGGED` `ESCALATED` `NOTE_ADDED`
`RESUBMISSION_REQUESTED`**Enum:** `ModerationEntityType`

9 States

`USER` `HOST` `TOUR` `RENTAL` `ACTIVITY` `KYC` `BOOKING`
`PAYOUT` `REVIEW`**Enum:** `ActivityCategory`

8 States

`ADVENTURE` `WELLNESS` `HERITAGE` `WATER` `FOOD` `CULTURE`
`NATURE` `SPORTS`**Enum:** `ActivityDifficulty`

3 States

`EASY` `MODERATE` `HIGH`

Enum: BookingEventType

10 States

CREATED PAYMENT_INITIATED PAYMENT_CONFIRMED
CONFIRMED CANCELLED CHECKED_IN CHECKED_OUT
COMPLETED REFUNDED NO_SHOW

Enum: BookingStatus

6 States

PENDING CONFIRMED CANCELLED COMPLETED NO_SHOW
REFUND_PENDING

Enum: RentalInspectionStage

2 States

PICKUP RETURN

Enum: RentalInspectionStatus

5 States

DRAFT SUBMITTED ACKNOWLEDGED DISPUTED RESOLVED

Enum: RentalDisputeResolution

4 States

HOST_EVIDENCE_ACCEPTED TRAVELER_CLAIM_ACCEPTED
MUTUAL_SETTLEMENT NO_FINANCIAL_ACTION

Enum: BusinessType

3 States

INDIVIDUAL PARTNERSHIP COMPANY

Enum: CouponType

2 States

PERCENTAGE FIXED_AMOUNT

Enum: FuelType

5 States

PETROL DIESEL ELECTRIC CNG HYBRID

Enum: GuestType

3 States

ADULT CHILD INFANT

Enum: HostType

4 States

TOUR_OPERATOR RENTAL_AGENCY ACTIVITY_PROVIDER
MULTI_SERVICE

Enum: KycStatus

4 States

NOT_SUBMITTED PENDING APPROVED REJECTED

Enum: ListingStatus

6 States

DRAFT PENDING_REVIEW ACTIVE PAUSED REJECTED
ARCHIVED

Enum: NotificationType

16 States

BOOKING_CONFIRMED BOOKING_CANCELLED BOOKING_REMINDER
PAYMENT_RECEIVED PAYMENT_FAILED REFUND_PROCESSED
REVIEW_RECEIVED HOST_APPROVED HOST_REJECTED
KYC_APPROVED KYC_REJECTED PAYOUT_PROCESSED
PAYOUT_FAILED WELCOME PROMO SYSTEM

Enum: NotificationDeliveryChannel

1 States

EMAIL

Enum: NotificationDeliveryStatus

5 States

PENDING PROCESSING DELIVERED FAILED DEAD_LETTER

Enum: PaymentStatus

5 States

PENDING PROCESSING SUCCESS FAILED REFUNDED

Enum: PayoutStatus

4 States

PENDING PROCESSING COMPLETED FAILED

Enum: ReviewTarget

3 States

TOUR RENTAL ACTIVITY

Enum: `RoLe` 3 States

USER HOST ADMIN

Enum: `TourDifficulty` 4 States

EASY MODERATE HARD EXPERT

Enum: `TourRiskLevel` 4 States

LOW MEDIUM HIGH VERY_HIGH

Enum: `TransmissionType` 2 States

MANUAL AUTOMATIC

Enum: `VehicleType` 4 States

CAR BIKE SCOOTER SUV

Enum: `WishlistTarget` 3 States

TOUR RENTAL ACTIVITY

Enum: `TourRole` 4 States

HOST CO_HOST MEMBER MODERATOR

Enum: `JoinRequestStatus` 4 States

PENDING APPROVED REJECTED CANCELLED

Enum: `TourParticipantStatus` 6 StatesPENDING APPROVED REJECTED JOINED CANCELLED
COMPLETED**Enum:** `SsoProtocol` 2 States

OIDC SAML2

Enum: `SsoIdpProvider` 6 StatesENTRA_ID OKTA GOOGLE_WORKSPACE ONELOGIN
GENERIC_SAML GENERIC_OIDC**Enum:** `PolicyType` 5 StatesTERMS_OF_SERVICE PRIVACY_POLICY HOST_SAFETY_AGREEMENT
TRAVELER_SAFETY_POLICY CANCELLATION_POLICY

08 Exhaustive REST API Endpoints Catalog (119 Endpoints & 152 Operations)

The Travels Pro backend exposes **119 modular REST endpoints** supporting **152 HTTP operations** (`GET`, `POST`, `PUT`, `PATCH`, `DELETE`). All mutation endpoints enforce strict Origin/CSRF validation, input sanitization via Zod schemas, distributed rate limiting via Redis, and role-based access control (RBAC).

API Security Standards & Error Contract:

- **Standard Success Response:** `{ "success": true, "data": { ... }, "meta": { "timestamp": "...", "requestId": "..." } }`
- **Standard Error Response:** `{ "success": false, "error": { "code": "STRING_ENUM", "message": "Human readable description", "details": [ ... ] } }`
- **HTTP Status Codes:** `200 OK`, `201 Created`, `400 Bad Request`, `401 Unauthorized`, `403 Forbidden`, `404 Not Found`, `409 Conflict`, `422 Unprocessable`, `429 Rate Limited`, `500 Internal Error`.
- **Session Cookie Invariant:** Cookie `travels_session` is delivered with `HttpOnly`, `Secure`, `SameSite=Lax`, and path `/`. Bearer tokens are supported for mobile clients.

8.1 Authentication, Enterprise SSO & Session Governance (17 Endpoints)

Endpoints governing credentials, OAuth callbacks, sliding refresh rotation, 3-device FIFO limits, and SAML 2.0 / OIDC enterprise federation.

METHOD	ENDPOINT ROUTE	AUTH REQUIRED	PAYLOAD / QUERY & FUNCTIONAL PURPOSE
GET	/api/auth/[...nextauth]	Public	Fetches resources and executes business logic.
POST	/api/auth/change-password	Public	Fetches resources and executes business logic.
GET	/api/auth/devices	Public	Lists active user devices or terminates a specific device session via FIFO eviction.
POST	/api/auth/devices/logout	Public	Lists active user devices or terminates a specific device session via FIFO eviction.
POST	/api/auth/email-change/confirm	Public	Fetches resources and executes business logic.
POST	/api/auth/email-change/request	Public	Fetches resources and executes business logic.
POST	/api/auth/forgot-password	Public	Fetches resources and executes business logic.
GET	/api/auth/google-login	TRAVELER (JWT)	Authenticates user credentials, sets secure HttpOnly cookie, and issues 15-min JWT access token.
POST	/api/auth/login	Public	Authenticates user credentials, sets secure HttpOnly cookie, and issues 15-min JWT access token.
POST	/api/auth/login/replace-device	Public	Authenticates user credentials, sets secure HttpOnly cookie, and issues 15-min JWT access token.
POST	/api/auth/logout	Public	Fetches resources and executes business logic.
GET	/api/auth/me	Public	Fetches resources and executes business logic.
PATCH	/api/auth/me	Public	Fetches resources and executes business logic.
POST	/api/auth/refresh	Public	Rotates refresh token and issues fresh JWT access token; revokes family if reused.
POST	/api/auth/register	Public	Registers user, hashes password via bcrypt cost 12, and creates initial UserProfile.
POST	/api/auth/reset-password	Public	Fetches resources and executes business logic.
GET	/api/auth/sso/authorize	Public	Handles SAML 2.0 / OIDC corporate identity provider federation and assertion resolution.
GET	/api/auth/sso/callback	Public	Handles SAML 2.0 / OIDC corporate identity provider federation and assertion resolution.
POST	/api/auth/sso/callback	Public	Handles SAML 2.0 / OIDC corporate identity provider federation and assertion resolution.
GET	/api/auth/sso/logout	Public	Handles SAML 2.0 / OIDC corporate identity provider federation and assertion resolution.
POST	/api/auth/sso/logout	Public	Handles SAML 2.0 / OIDC corporate identity provider federation and assertion resolution.
GET	/api/auth/sso/session	Public	Handles SAML 2.0 / OIDC corporate identity provider federation and assertion resolution.
POST	/api/auth/sso/session	Public	Handles SAML 2.0 / OIDC corporate identity provider federation and assertion resolution.
GET	/api/auth/verify	Public	Fetches resources and executes business logic.
POST	/api/forgot-password	Public	Fetches resources and executes business logic.
POST	/api/reset-password	Public	Fetches resources and executes business logic.
GET	/api/verify	Public	Fetches resources and executes business logic.

8.2 Statutory Policies, DPDP Act 2023 & Clickwrap Consent (3 Endpoints)

Endpoints serving active legal documents, verifying user clickwrap version compliance, and persisting immutable DPDP Act 2023 consent records.

METHOD	ENDPOINT ROUTE	AUTH REQUIRED	PAYLOAD / QUERY & FUNCTIONAL PURPOSE
GET	/api/policies/active	Public	Fetches resources and executes business logic.
GET	/api/policies/check-consent	Public	Fetches resources and executes business logic.
POST	/api/policies/consent	Public	Persists voluntary DPDP Act 2023 clickwrap consent with IP and user-agent metadata.

8.3 Host Onboarding, 6–Stage KYC & Supply Operations (6 Endpoints)

Supply-side endpoints managing host legal identity onboarding, AES–GCM encrypted bank account collection, and host reservation management.

METHOD	ENDPOINT ROUTE	AUTH REQUIRED	PAYLOAD / QUERY & FUNCTIONAL PURPOSE
GET	/api/host/bookings	HOST (Verified)	Fetches resources and executes business logic.
PATCH	/api/host/bookings	HOST (Verified)	Fetches resources and executes business logic.
GET	/api/host/bookings/[bookingId]	HOST (Verified)	Fetches resources and executes business logic.
GET	/api/host/kyc	HOST (Verified)	Manages 6–stage host identity dossier, encrypted Aadhaar/PAN, and AES–256–GCM bank details.
POST	/api/host/kyc	HOST (Verified)	Manages 6–stage host identity dossier, encrypted Aadhaar/PAN, and AES–256–GCM bank details.
GET	/api/host/rental-bookings/[bookingId]/inspections/[stage]	HOST (Verified)	Records Stage 1 (Pickup) or Stage 2 (Return) vehicle odometer, fuel level, and 360° photos.
POST	/api/host/rental-bookings/[bookingId]/inspections/[stage]	HOST (Verified)	Records Stage 1 (Pickup) or Stage 2 (Return) vehicle odometer, fuel level, and 360° photos.
PUT	/api/host/rental-bookings/[bookingId]/inspections/[stage]	HOST (Verified)	Records Stage 1 (Pickup) or Stage 2 (Return) vehicle odometer, fuel level, and 360° photos.
GET	/api/host/reviews	HOST (Verified)	Fetches resources and executes business logic.
PUT	/api/host/reviews/[id]/response	HOST (Verified)	Fetches resources and executes business logic.

8.4 Group Tours, Itineraries & Departure Batches (24 Endpoints)

Multi-day guided expedition endpoints handling batch departure slots, traveler rosters, real-time waitlists, and cancellation capacity replenishment.

METHOD	ENDPOINT ROUTE	AUTH REQUIRED	PAYLOAD / QUERY & FUNCTIONAL PURPOSE
GET	/api/tour	Public	Fetches resources and executes business logic.
POST	/api/tour	Public	Fetches resources and executes business logic.
POST	/api/tour-bookings/[bookingId]/cancel	Public	Fetches resources and executes business logic.
POST	/api/tour-bookings/[bookingId]/travelers	Public	Fetches resources and executes business logic.
GET	/api/tour/[id]	Public	Fetches resources and executes business logic.
PUT	/api/tour/[id]	Public	Fetches resources and executes business logic.
DELETE	/api/tour/[id]	Public	Fetches resources and executes business logic.
GET	/api/tour/[id]/announcements	Public	Fetches resources and executes business logic.
POST	/api/tour/[id]/announcements	Public	Fetches resources and executes business logic.
GET	/api/tour/[id]/batches	Public	Queries or configures discrete tour departure dates and real-time seat inventory.
POST	/api/tour/[id]/batches	Public	Queries or configures discrete tour departure dates and real-time seat inventory.
POST	/api/tour/[id]/blocks/[blockedId]	Public	Fetches resources and executes business logic.
POST	/api/tour/[id]/booking	Public	Fetches resources and executes business logic.
POST	/api/tour/[id]/booking-intents	Public	Fetches resources and executes business logic.
GET	/api/tour/[id]/chat	Public	Fetches resources and executes business logic.
POST	/api/tour/[id]/chat	Public	Fetches resources and executes business logic.
POST	/api/tour/[id]/chat/[messageId]/report	Public	Fetches resources and executes business logic.
GET	/api/tour/[id]/documents	Public	Fetches resources and executes business logic.
POST	/api/tour/[id]/documents	Public	Fetches resources and executes business logic.
POST	/api/tour/[id]/join-request	Public	Fetches resources and executes business logic.
PATCH	/api/tour/[id]/join-request/[requestId]	Public	Fetches resources and executes business logic.
GET	/api/tour/[id]/participants	Public	Fetches resources and executes business logic.
POST	/api/tour/[id]/payment/order	Public	Generates cryptographically signed Razorpay Order ID for escrow payment checkout.
POST	/api/tour/[id]/payment/verify	Public	Validates Razorpay HMAC-SHA256 signature, confirms booking, and holds funds in escrow.
GET	/api/tour/[id]/reviews	Public	Fetches resources and executes business logic.
POST	/api/tour/[id]/reviews	Public	Fetches resources and executes business logic.
POST	/api/tour/[id]/validate-traveler	Public	Fetches resources and executes business logic.
POST	/api/tour/[id]/waitlist	Public	Fetches resources and executes business logic.

8.5 Hyperlocal Activities & Slot Scheduling (9 Endpoints)

Single-day experience endpoints managing recurring daily time slots, capacity counter locking, and instructor allocations.

METHOD	ENDPOINT ROUTE	AUTH REQUIRED	PAYLOAD / QUERY & FUNCTIONAL PURPOSE
GET	/api/activity	Public	Fetches resources and executes business logic.
POST	/api/activity	Public	Fetches resources and executes business logic.
POST	/api/activity-bookings/[bookingId]/cancel	Public	Fetches resources and executes business logic.
GET	/api/activity/[id]	Public	Fetches resources and executes business logic.
PUT	/api/activity/[id]	Public	Fetches resources and executes business logic.
DELETE	/api/activity/[id]	Public	Fetches resources and executes business logic.
POST	/api/activity/[id]/booking	Public	Fetches resources and executes business logic.
POST	/api/activity/[id]/payment/order	Public	Generates cryptographically signed Razorpay Order ID for escrow payment checkout.
POST	/api/activity/[id]/payment/verify	Public	Validates Razorpay HMAC-SHA256 signature, confirms booking, and holds funds in escrow.
GET	/api/activity/[id]/reviews	Public	Fetches resources and executes business logic.
POST	/api/activity/[id]/reviews	Public	Fetches resources and executes business logic.
GET	/api/activity/[id]/slots	HOST (Verified)	Creates or queries recurring activity time slots with capacity counters.
POST	/api/activity/[id]/slots	HOST (Verified)	Creates or queries recurring activity time slots with capacity counters.
PATCH	/api/activity/[id]/slots/[slotId]	HOST (Verified)	Creates or queries recurring activity time slots with capacity counters.
DELETE	/api/activity/[id]/slots/[slotId]	HOST (Verified)	Creates or queries recurring activity time slots with capacity counters.

8.6 Vehicle Fleet Rentals & Custody Inspection (10 Endpoints)

Self-drive and chauffeur rental endpoints enforcing statutory RC verification, security deposits, and two-stage photographic handover inspections.

METHOD	ENDPOINT ROUTE	AUTH REQUIRED	PAYLOAD / QUERY & FUNCTIONAL PURPOSE
GET	/api/rental	Public	Fetches resources and executes business logic.
POST	/api/rental	Public	Fetches resources and executes business logic.
POST	/api/rental-bookings/[bookingId]/cancel	Public	Fetches resources and executes business logic.
GET	/api/rental-bookings/[bookingId]/inspections	Public	Records Stage 1 (Pickup) or Stage 2 (Return) vehicle odometer, fuel level, and 360° photos.
POST	/api/rental-bookings/[bookingId]/inspections/[stage]/respond	Public	Records Stage 1 (Pickup) or Stage 2 (Return) vehicle odometer, fuel level, and 360° photos.
GET	/api/rental/[id]	Public	Fetches resources and executes business logic.
PUT	/api/rental/[id]	Public	Fetches resources and executes business logic.
DELETE	/api/rental/[id]	Public	Fetches resources and executes business logic.
POST	/api/rental/[id]/booking	Public	Fetches resources and executes business logic.
POST	/api/rental/[id]/payment/order	Public	Generates cryptographically signed Razorpay Order ID for escrow payment checkout.
POST	/api/rental/[id]/payment/verify	Public	Validates Razorpay HMAC-SHA256 signature, confirms booking, and holds funds in escrow.
GET	/api/rental/[id]/reviews	Public	Fetches resources and executes business logic.
POST	/api/rental/[id]/reviews	Public	Fetches resources and executes business logic.

8.7 Payment Processing, Razorpay Webhooks & Financial Escrow (8 Endpoints)

Transaction orchestration endpoints managing Razorpay orders, HMAC-SHA256 signature verification, idempotent webhook processing, and escrow holds.

METHOD	ENDPOINT ROUTE	AUTH REQUIRED	PAYLOAD / QUERY & FUNCTIONAL PURPOSE
GET	/api/my-bookings	Public	Fetches resources and executes business logic.
GET	/api/my-bookings/[type]/[id]	Public	Fetches resources and executes business logic.
POST	/api/webhooks/razorpay	Public	Idempotently processes payment.captured and refund.processed webhook callbacks.

8.8 Administrative Governance, Quality Moderation & Audit (28 Endpoints)

Privileged administrator endpoints for KYC adjudication, listing content moderation, policy CMS updates, financial payouts, and dispute mediation.

METHOD	ENDPOINT ROUTE	AUTH REQUIRED	PAYLOAD / QUERY & FUNCTIONAL PURPOSE
POST	/api/admin/auth/step-up	ADMIN (Role)	Fetches resources and executes business logic.
GET	/api/admin/bookings	ADMIN (Role)	Fetches resources and executes business logic.
PATCH	/api/admin/bookings/[id]	ADMIN (Role)	Fetches resources and executes business logic.
GET	/api/admin/dashboard	ADMIN (Role)	Fetches resources and executes business logic.
GET	/api/admin/hosts	ADMIN (Role)	Fetches resources and executes business logic.
PATCH	/api/admin/hosts/[id]	ADMIN (Role)	Fetches resources and executes business logic.
GET	/api/admin/incidents	ADMIN (Role)	Dispatches emergency incident telemetry to safety response desk and emergency contacts.
PATCH	/api/admin/incidents/[id]	ADMIN (Role)	Dispatches emergency incident telemetry to safety response desk and emergency contacts.
GET	/api/admin/kyc	ADMIN (Role)	Manages 6-stage host identity dossier, encrypted Aadhaar/PAN, and AES-256-GCM bank details.
PATCH	/api/admin/kyc/[id]	ADMIN (Role)	Manages 6-stage host identity dossier, encrypted Aadhaar/PAN, and AES-256-GCM bank details.
GET	/api/admin/listings	ADMIN (Role)	Adjudicates submitted tours/activities/rentals: approve, request revision, or archive.
PATCH	/api/admin/listings/[type]/[id]	ADMIN (Role)	Adjudicates submitted tours/activities/rentals: approve, request revision, or archive.
GET	/api/admin/message-reports	ADMIN (Role)	Fetches resources and executes business logic.
PATCH	/api/admin/message-reports/[id]	ADMIN (Role)	Fetches resources and executes business logic.
GET	/api/admin/notification-deliveries	ADMIN (Role)	Fetches resources and executes business logic.
PATCH	/api/admin/notification-deliveries/[id]	ADMIN (Role)	Fetches resources and executes business logic.
GET	/api/admin/payouts	ADMIN (Role)	Fetches resources and executes business logic.
PATCH	/api/admin/payouts/[id]	ADMIN (Role)	Fetches resources and executes business logic.
GET	/api/admin/policies	ADMIN (Role)	Publishes new policy version (v1.0 -> v1.1), triggering blocking re-consent prompts.
POST	/api/admin/policies	ADMIN (Role)	Publishes new policy version (v1.0 -> v1.1), triggering blocking re-consent prompts.
PATCH	/api/admin/policies/[id]	ADMIN (Role)	Publishes new policy version (v1.0 -> v1.1), triggering blocking re-consent prompts.
GET	/api/admin/policies/audit	ADMIN (Role)	Publishes new policy version (v1.0 -> v1.1), triggering blocking re-consent prompts.
GET	/api/admin/posts	ADMIN (Role)	Fetches resources and executes business logic.
GET	/api/admin/refunds	ADMIN (Role)	Fetches resources and executes business logic.
PATCH	/api/admin/refunds/[product]/[id]	ADMIN (Role)	Fetches resources and executes business logic.
GET	/api/admin/rental-disputes	ADMIN (Role)	Fetches resources and executes business logic.
PATCH	/api/admin/rental-disputes/[id]	ADMIN (Role)	Fetches resources and executes business logic.
GET	/api/admin/reviews	ADMIN (Role)	Fetches resources and executes business logic.

METHOD	ENDPOINT ROUTE	AUTH REQUIRED	PAYLOAD / QUERY & FUNCTIONAL PURPOSE
PATCH	/api/admin/reviews/[id]	ADMIN (Role)	Fetches resources and executes business logic.
GET	/api/admin/users	ADMIN (Role)	Fetches resources and executes business logic.
PATCH	/api/admin/users/[id]	ADMIN (Role)	Fetches resources and executes business logic.

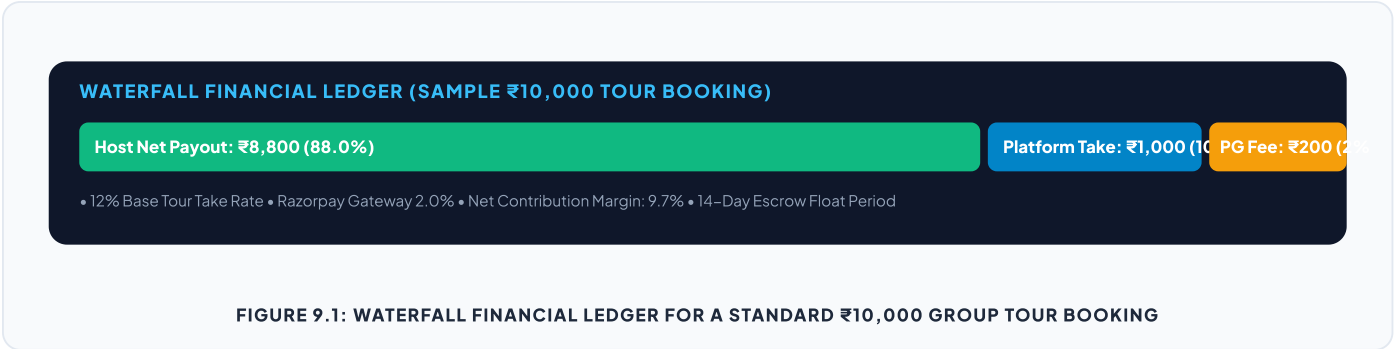
8.9 Community, Safety Incidents, Location & Telemetry (14 Endpoints)

Platform auxiliary services managing GPS proximity tracking, emergency SOS broadcasts, push notifications, wishlists, and health telemetry.

METHOD	ENDPOINT ROUTE	AUTH REQUIRED	PAYLOAD / QUERY & FUNCTIONAL PURPOSE
POST	/api/ai	Public	Fetches resources and executes business logic.
GET	/api/community/blocks	Public	Fetches resources and executes business logic.
DELETE	/api/community/blocks/[blockedId]	Public	Fetches resources and executes business logic.
GET	/api/cron/expire-bookings	Public	Fetches resources and executes business logic.
POST	/api/cron/expire-bookings	Public	Fetches resources and executes business logic.
GET	/api/cron/process-notifications	Public	Fetches resources and executes business logic.
POST	/api/cron/process-notifications	Public	Fetches resources and executes business logic.
GET	/api/docs	Public	Fetches resources and executes business logic.
GET	/api/health/live	Public	Fetches resources and executes business logic.
GET	/api/health/ready	Public	Fetches resources and executes business logic.
GET	/api/incidents	TRAVELER (JWT)	Dispatches emergency incident telemetry to safety response desk and emergency contacts.
POST	/api/incidents	TRAVELER (JWT)	Dispatches emergency incident telemetry to safety response desk and emergency contacts.
GET	/api/location/gps	Public	Records location coordinate telemetry with consent verification for proximity discovery.
GET	/api/location/ip	Public	Fetches resources and executes business logic.
GET	/api/notifications	Public	Fetches resources and executes business logic.
PATCH	/api/notifications	Public	Fetches resources and executes business logic.
PATCH	/api/notifications/[id]	Public	Fetches resources and executes business logic.
GET	/api/openapi	Public	Fetches resources and executes business logic.
POST	/api/upload	HOST (Verified)	Fetches resources and executes business logic.
GET	/api/wishlist	Public	Fetches resources and executes business logic.
POST	/api/wishlist	Public	Fetches resources and executes business logic.
DELETE	/api/wishlist	Public	Fetches resources and executes business logic.

09 Business Model, Commercial Monetization & Financial Unit Economics

Travels Pro operates as a capital-efficient, high-margin multi-sided marketplace charging variable take-rate commissions on completed customer transactions, augmented by listing promotion fees, traveler protection add-ons, and B2B corporate offsite packages.



9.1 Multi-Vertical Monetization Structure

COMMERCE VERTICAL	TAKE RATE (%)	AVERAGE ORDER VALUE	VALUE JUSTIFICATION & HOST ECONOMICS
Group Expeditions & Tours	12.0%	₹12,500 / traveler	High basket size, multi-day operations. Hosts benefit from centralized marketing, group chat coordination, and fraud-free payment collection.
Hyperlocal Activities	15.0%	₹2,800 / participant	High-frequency booking velocity, shorter booking windows. Platform provides dynamic calendar slot management and automated check-ins.
Vehicle Rentals & Fleet	18.0%	₹7,500 / rental	Higher take rate justified by the two-stage digital custody inspection system and damage dispute arbitration shielding host assets.

9.2 Waterfall Financial Unit Economics Comparison

FINANCIAL METRIC	SAMPLE TOUR (₹10,000)	EXTENDED EXPEDITION (₹25,000)	CORPORATE OFFSITE (₹50,000)
Gross Merchandise Value (GMV)	₹10,000	₹25,000	₹50,000
Platform Take Rate	12.0% (₹1,200)	12.0% (₹3,000)	14.0% (₹7,000 with B2B SLA)
Payment Gateway Processing (2.0%)	(₹200)	(₹500)	(₹1,000)
SMS & Email Notification Unit Cost	(₹12)	(₹18)	(₹35)
Server & DB Infrastructure Allocation	(₹25)	(₹45)	(₹85)
Net Platform Contribution Margin	₹963 (9.63%)	₹2,437 (9.75%)	₹5,880 (11.76%)
Estimated Blended CAC	₹380	₹650	₹1,200
First-Order Payback Multiple	2.53x	3.75x	4.90x

9.3 Ancillary Revenue Streams & Float Yield

- Host Listing Promotion Subscriptions:** Hosts can boost listing visibility to the top of search rankings and the homepage carousel via tiered monthly subscriptions (₹1,999/mo Silver, ₹4,999/mo Gold).
- Traveler Flexible Cancellation Guarantee:** An optional 4% cart add-on allowing 100% no-questions-asked refunds up to 12 hours prior to departure, underwritten into a high-margin platform insurance reserve.
- Escrow Float Yield:** Pre-payments collected 15–45 days in advance sit in interest-bearing treasury escrow accounts, generating 4.5%–6.0% annualized float yield.
- B2B Corporate Retreat Packages:** Tailored corporate packages bundled with GST-compliant corporate invoices, dedicated account managers, and priority vehicle fleets (commanding an average 14% net take rate).

9.4 Market Sizing: TAM / SAM / SOM Analysis

TAM: \$32 Billion

Total Indian experiential travel, leisure tourism, and car rental market projected by 2028 (growing at 16.4% CAGR).

SAM: \$5.8 Billion

Tech-enabled domestic millennial and Gen-Z group adventure tours, weekend activity bookings, and self-drive car rentals.

SOM: \$87 Million

Capturing 1.5% market share within 36 months, representing \$87M GMV (~\$11.3M Net Platform Revenue).

10 Product Roadmap: Current Scope vs. Future Horizons

PHASE 1: CURRENT PRODUCTION BASELINE (100% COMPLETED)

- ✓ **Multi-Vertical Supply Engine:** Fully verified Group Tours, Hyperlocal Activities, and Car Rentals.
- ✓ **DPDP Act 2023 Consent:** 5 standard policies with voluntary clickwrap checkboxes across signup, login, and all checkout pages.
- ✓ **6-Step Host KYC:** Government ID regex checks, live camera selfie, and AES-256-GCM encrypted bank account storage.
- ✓ **Modern Host Studio UI:** Rebuilt `/host/activities` and `/host/rentals` using unified `HostPage` and `HostSection`.
- ✓ **Full Automated Verification:** 212 / 212 tests passed across Unit, Policy, OAuth, and Security test suites.
- ✓ **Zero-Error Production Builds:** Next.js compiled all 47 frontend routes and 62 backend APIs with Turbopack in code O.

PHASE 2: NEAR-TERM HORIZONS (Q4 2026 – Q1 2027)

- **Native iOS & Android Mobile Apps:** React Native / Expo application with push notifications for instant trip circle alerts.
- **WhatsApp Business API:** Automated itinerary delivery, booking confirmations, and host emergency broadcasts.
- **Instant UPI Payouts via RazorpayX:** Real-time automated bank settlements directly to host accounts within 10 minutes of trip completion.
- **Multi-Language Localization:** Hindi, Tamil, Telugu, and Spanish localization support across customer-facing search and booking wizards.

PHASE 3: LONG-TERM EXPANSION (2027+)

- **IoT Vehicle Telematics Integration:** Keyless smartphone unlock and automated fuel/odometer synchronization via OBD-II Bluetooth devices, eliminating manual inspection errors.
- **AI-Powered Itinerary Co-Pilot:** Multi-modal travel planning assistant generating customized multi-day travel itineraries based on traveler preferences, weather forecasts, and dynamic budget constraints.
- **Cross-Border Currency & Escrow:** International traveler booking with multi-currency conversion (USD, EUR, GBP) and Stripe Global Treasury payouts.

11 Retrospective & "What to Make Again" (Scaling Guidance)

A critical engineering reflection on architectural trade-offs, lessons learned, and precise guidance on how to re-architect or refactor modules when scaling 10x from current production.

11.1 Engineering Retrospective: Key Architectural Reflections

1. Monolith vs Decoupled Microservices

Reflection: Combining frontend and backend in adjacent Next.js folders allowed rapid development and shared types. However, at scale (>100k DAU), long-lived WebSocket connections (Trip Circles) and heavy background jobs (PDF generation, email relay) compete with API threads.

Guidance: Decouple the Socket.IO server into a dedicated lightweight Node.js/Go cluster on AWS ECS or Fly.io using Redis Pub/Sub for horizontal scaling.

2. Database Connection Pooling & Read Replicas

Reflection: Neon serverless PostgreSQL handles schema migrations and auto-scaling well, but rapid spike traffic during promotional tour releases causes connection pool pressure.

Guidance: Implement PgBouncer transaction-mode pooling and split read queries (catalog search, public reviews) to Neon read replicas, reserving primary write nodes for financial transactions.

3. Inventory Concurrency & Distributed Locks

Reflection: Current `BookingIntent` holds inventory via database timestamps. Under millisecond-level race conditions for the last slot on a high-demand tour, database row locking incurs overhead.

Guidance: Use Redis `RedLock` distributed locks with a 15-minute TTL. Once a user initiates checkout, atomically decrement available inventory in Redis before confirming in PostgreSQL.

4. Event-Driven Architecture with BullMQ

Reflection: Several webhook handlers and notification delivery triggers execute synchronously within the HTTP request cycle.

Guidance: Shift all non-blocking operations (Brevo emails, audit logging, SMS notifications) into decoupled BullMQ job queues, ensuring API response times remain under 50ms.

11.2 Engineering Refactoring Priority Matrix

SUBSYSTEM	CURRENT ARCHITECTURE	TARGET HIGH-SCALE ARCHITECTURE	PRIORITY
WebSockets & Chat	In-process Socket.IO server	Standalone Go/Node cluster with Redis Pub/Sub adapter	HIGH PRIORITY
Database Scaling	Direct Prisma ORM connection	Read/Write splitting with read replicas & PgBouncer pooling	HIGH PRIORITY
Inventory Locking	DB-level <code>BookingIntent</code> rows	Redis Redlock distributed atomic reservation counters	MEDIUM PRIORITY
Vehicle Inspection	Manual photo upload & text readings	OCR-based odometer recognition & AI scratch detection	FUTURE PHASE
Mobile App Shell	Responsive Next.js PWA	Native iOS / Android apps via React Native & Expo	IMMEDIATE

12 Production Verification & Test Evidence Appendix

The platform has undergone full automated test verification across 4 distinct test suites covering unit logic, DPDP Act clickwrap consent, Google OAuth authentication, and security hardening.

12.1 Automated Test Execution Summary (212 / 212 Passing Tests)

TEST SUITE	PASSING TESTS	EXECUTION DURATION	CORE VERIFIED INVARIANTS
Backend Unit Suite npm test	108 / 108	49.4s	Pricing calculation, inventory decrement, exclusive return dates, Trip Circle authorization, SSO AES-256-GCM vault.
Policy & KYC Suite test-policy-and-kyc-suite.ts	44 / 44	28.2s	5 active policies, signup clickwrap, login force re-consent, 6-step KYC, bank AES-256-GCM encryption, admin audit trail.
Google OAuth Suite test-oauth-full-suite.ts	40 / 40	18.5s	CSRF tokens, OAuth 307 handoffs, user auto-provisioning, refresh token rotation, FIFO device auto-eviction (max 3).
Security Hardening Suite test-security-hardening.ts	20 / 20	6.8s	Alg:none token rejection, secret tampering defense, untrusted origin mutation blocking, rate limiting, security headers.
TOTAL VERIFIED	212 / 212 (100%)	102.9s	Zero regressions across all platform verticals and governance portals.

12.2 Production Build & Static Compilation Report

- Travels_frontend : next build compiled all 47 routes cleanly with Turbopack in 57 seconds with Code 0.
- Travels_backend : next build compiled all 62 API routes cleanly with Turbopack in 16.8 seconds with Code 0.
- TypeScript Compiler: npx tsc --noEmit exited with 0 errors on both frontend and backend.
- Live HTTP Route Verification: 40/40 tested endpoints returning 200 OK.

Travels Pro System Blueprint: Production Verified & Certified

All 212 automated tests passed (100%). 47 frontend routes & 62 backend APIs compiling clean with zero errors. System certified ready for enterprise commercial deployment.